

Healthcare Pathway Description Standard (HPDS), Release 1

Health informatics: Healthcare pathway description: Catalogue-ready business artefacts, conceptual model and conformance

Working Draft 0.5

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Status of this document

This is HPDS Release 1, Working Draft 0.5 (document identifier AHI_HPDS_R1_WD05_2026SEP), prepared by Agile Health Informatics Ltd. It cancels and replaces Working Draft 0.4. Its status is working draft for review; its content is unapproved.

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Version	Date	Summary of changes
WD 0.1	–	First working draft.
WD 0.2	–	Catalogue-centred scope; four conformance levels; authority separated from lifecycle and intended use.
WD 0.3	2026-09-17	Structured data types, relationship attributes and cardinalities, sub-pathway inclusion and validation rules restored.
WD 0.4	2026-09-26	Formal representations; unordered stages; accountableFor; pathway-level relationships at Level 1; rights; Task withdrawn.
WD 0.5	2026-09-27	HL7 document format. Consistency and implementability revision: descriptive behavioural semantics; determinate graph validation; governance events; definition deviations; implementation clinical authority; expression concepts; Level 4 coverage; representation discrepancies; extension handling; obligations by level; release package. See Appendix L.

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The editor acknowledges the authors of the HL7 Field Guide to Shareable Clinical Pathways, whose illustrative "Hello Patient" models are used, with attribution, as the source of the sequential example in Appendix I. Contributors to the review of this working draft are to be recorded here as review proceeds.

Table of contents

Status of this document	2
Document information	3
Revision history	4
Acknowledgements	5
Executive summary	9
1 Introduction	10
1.1 Purpose	10
1.2 Scope	10
1.3 Audience	11
1.4 Definition, implementation, instance and catalogue	11
1.5 Relationship with other standards	11
1.6 Conformance approach	12
1.7 Authority and intended use	12
1.8 Conventions	12
1.9 Implementation readiness	12
2 Normative references	13
3 Terms and definitions	14
4 Abbreviated terms	19
5 Conformance	20
5.1 Conformance classes	20
5.2 Pathway definition conformance levels	20
5.3 Minimum pathway definition dataset	20
5.4 Pathway implementation conformance	21
5.5 Pathway catalogue conformance	21
5.6 Producer and consumer conformance	21
5.7 Obligations by conformance level	22
6 Conceptual framework	23
6.1 Separation of definition, implementation and instance	23
6.2 Design principles	23
6.3 Level of detail	24
7 Catalogue model	25
7.1 PathwayCatalogue	25
7.2 CatalogueEntry	25
7.3 Catalogue discovery	25
8 Pathway conceptual model	26
8.1 General	26
8.2 Data types	26
8.3 Common attributes	27
8.4 HealthcarePathway	27
8.5 Population	28
8.6 EntryPoint	28
8.7 ExitPoint	29
8.8 PathwayStage	29
8.9 Activity	30
8.10 FormalRepresentation	30
8.11 DecisionPoint	32
8.12 PathwayBranch	32
8.13 Event	32
8.14 Role	33
8.15 Service	33
8.16 CareSetting	33
8.17 InformationRequirement	33
8.18 InformationProduct	33
8.19 ClinicalConcept	34
8.20 Outcome	34

8.21 Measure	34
8.22 Constraint	34
8.23 Exception	35
8.24 EvidenceSource	35
8.25 InteroperabilityRequirement	35
8.26 Structured data types	36
8.27 PathwayImplementation	38
8.28 Implementation data types	39
8.29 Reference points for pathway instances	40
9 Relationships and graph semantics	41
9.1 Relationship representation	41
9.2 Relationship types	41
9.3 Structural containment and inclusion	42
9.4 Behavioural patterns	42
9.5 Iteration	43
9.6 Behavioural reachability	43
9.7 Process execution	44
9.8 Descriptive behavioural semantics	44
9.9 Graph validation outcomes	45
10 Temporal representation	46
10.1 TemporalConstraint	46
10.2 Relative time	46
10.3 Consistency	46
11 Semantic binding	47
11.1 General	47
11.2 Binding requirements	47
11.3 Local terms	47
11.4 Change in bound resources	47
11.5 Binding report	47
12 Interoperability requirements	48
13 Specialisation, variation and profiles	49
13.1 Specialisation	49
13.2 Inheritance and deviation	49
13.3 Domain profiles	49
13.4 Expression profiles	49
13.5 Profile identification	49
13.6 Extensions	50
13.7 Expression coverage at Level 4	50
14 Governance, provenance and lifecycle	52
14.1 Governance metadata	52
14.2 Content immutability and governance state	52
14.3 Provenance	52
14.4 Change history	52
14.5 Lifecycle	52
14.6 Historical and current state	53
15 Identification and versioning	54
15.1 Identifiers	54
15.2 Stability across versions	54
15.3 Version identifiers	54
15.4 References between descriptions	54
16 Authority and intended use	55
16.1 Authority status	55
16.2 Intended use	55
16.3 Automated use	55
17 Human-readable catalogue representation	56
18 Machine-readable representation	57
19 Serialisation	58
19.1 Normative JSON serialisation	58

19.2 Other serialisations	58
19.3 Authority of the conceptual model	58
Appendix A Validation rules	59
A.1 General	59
Appendix B JSON serialisation	62
B.1 General	62
B.2 Serialisation rules	62
B.3 Collections	62
Appendix C Mapping to HL7 FHIR	64
Appendix D Mapping to BPMN, CMMN and DMN and relationship with the HL7 Field Guide	65
D.1 General	65
D.2 Correspondence with BPMN, CMMN and DMN	65
D.3 Relationship with the HL7 Field Guide to Shareable Clinical Pathways	65
D.4 Decision	66
Appendix E Mapping to ArchiMate	67
Appendix F Relationship with ISO 13940:2026 and the standards landscape	68
F.1 ISO 13940:2026	68
F.2 WHO SMART Guidelines and Digital Adaptation Kits	68
F.3 Standards landscape	68
Appendix G Human-readable pathway catalogue record	69
Appendix H Mapping existing tabular catalogues	70
H.1 Method	70
H.2 Findings from the Frailty and Multimorbidity Neighbourhood Care Pathway Catalogue	70
Appendix I Examples, test cases and results	71
I.1 Status of worked examples	71
I.2 Test cases and results	71
Appendix J Conformance checklist	73
Appendix K Development, validation and publication	74
K.1 Development priorities following WD 0.5	74
K.2 Repository structure	74
K.3 Success criteria	74
K.4 Principal changes in WD 0.4	74
K.5 Decision register	75
K.6 Open issues	76
K.7 Earlier history	76
Appendix L Change log from WD 0.4	77
Appendix M Requirements traceability	79
M.1 Method	79
M.2 Revised and new requirements	79
M.3 Conformance statements and validation rules	80
Appendix N Unresolved issues and missing implementation evidence	86
N.1 Minimum release package	86
N.2 Prioritised unresolved issues	86
Bibliography	88

If page numbers are not shown, right-click the table and select Update Field.

Executive summary

Healthcare pathways are usually held as guidance documents, diagrams, spreadsheets, local procedures and system configuration. In those forms they are difficult to discover, compare, govern, specialise and relate to the services, information and technology that realise them. The Healthcare Pathway Description Standard (HPDS) specifies a technology-neutral representation of a healthcare pathway as a governed business artefact that can be held in a pathway catalogue.

HPDS defines a conceptual model for pathway definitions, pathway implementations and pathway catalogues; requirements for identification, governance, provenance, lifecycle, authority and intended use; explicit structural, behavioural, dependency and responsibility relationships with descriptive, analysable semantics; semantic binding; specialisation and deviation; references to formal representations expressed in BPMN, CMMN, DMN, HL7 FHIR and similar standards; four graded conformance levels with validation rules; and normative JSON and human-readable catalogue representations. It keeps the intended model of care separate from any formal model, local implementation and individual patient record, and it keeps technical conformance separate from clinical authority, intended use and lifecycle status.

Working Draft 0.5 is a consistency and implementability revision of Working Draft 0.4. It resolves eleven review findings about behavioural semantics, graph validation, named organisations, missing model structures, immutability, Level 4 coverage, formal representations, extensions, adoption burden, implementation readiness and cross-document consistency. It does not widen the scope of the standard. Appendix L lists the changes, Appendix M traces revised requirements to the model and validation rules, and Appendix N lists unresolved issues and missing implementation evidence.

1 Introduction

1.1 Purpose

This section is informative.

Healthcare organisations routinely maintain catalogues of applications, interfaces, services, data assets, standards and capabilities. Pathways themselves are commonly held in less structured forms, including guidance documents, diagrams, spreadsheets, local operating procedures and system configuration. Pathways have been the subject of sustained definitional work [12], but that work did not produce a representation of the pathway as governed information. As a result, pathways are difficult to discover, compare, govern, specialise, analyse and relate consistently to services, information requirements and technology.

This document specifies a technology-neutral business representation of a healthcare pathway suitable for maintenance in a pathway catalogue. The representation is designed to remain intelligible to clinicians, managers and architects while supporting structured machine processing. ISO/TC 215, Health informatics, is the committee within whose scope the subject matter most naturally falls; this document is prepared in the conventions of HL7 specifications to support review by health informatics standards communities.

1.2 Scope

This document specifies requirements for representing healthcare pathways as technology-neutral, governed business artefacts suitable for inclusion in healthcare pathway catalogues. It specifies:

- a) a conceptual model for pathway definitions, pathway implementations and pathway catalogues, comprising entities, attributes, structured data types, relationships, cardinalities and constraints;
- b) requirements for catalogue identification, discovery, governance, provenance, rights and lifecycle, including the recording of governance events after publication;
- c) representation of population, entry, exit, stages, activities, decisions, events, roles, services, care settings, information requirements, information products, outcomes, measures, constraints, exceptions and evidence;
- d) explicit structural, behavioural, dependency and responsibility relationships, including inclusion of sub-pathways and relationships between pathways, and the descriptive behavioural semantics of those relationships (Chapter 9);
- e) semantic binding to external terminologies, data specifications and information standards;
- f) derivation of technology-neutral interoperability requirements from pathway dependencies;
- g) specialisation of reference pathways and explicit recording of deviations;
- h) authority status and intended use;
- i) reference from a pathway definition to formal representations of it expressed in external process, case, decision or plan-definition standards, and the handling of discrepancies between them;
- j) graded conformance levels, validation rules and conformance claims;
- k) a normative JSON serialisation and a normative human-readable catalogue record.

Its provisions apply across primary, community, secondary, tertiary and specialist care, diagnostics, mental health, maternity, medicines, rehabilitation, prevention and screening, urgent and emergency care, elective care, long-term condition management, end-of-life care and interfaces with social care. They apply to pathways whose behaviour is largely sequential and to pathways whose behaviour is largely adaptive.

This document defines descriptive behavioural semantics: a declarative graph that states the intended order, concurrency, choice, repetition, waiting, interruption, escalation and transition between the behavioural nodes of a pathway, and the containment of those nodes in stages. The graph has enough meaning to support the analyses listed in 9.8: structural validation, comparison of pathways and variants, dependency and impact analysis, derivation of interoperability requirements and mapping to formal representations. It does not define execution semantics. Where operational precision is required, it is provided by a formal representation (8.10) or an implementation artefact (8.28.2) expressed in an external standard.

The following are outside the scope of this document:

- a) determination of clinical content or clinical appropriateness;
- b) representation of individual patient records or pathway instances, other than the reference points recommended in 8.29;
- c) design of messages, application programming interfaces, endpoints, transport protocols or integration products;
- d) execution semantics, including run-time instance state, token or case-state models, scheduling, work assignment, transaction and error handling, evaluation order of expressions, and the behaviour of workflow engines, clinical decision support systems or orchestration platforms;
- e) the definition of computable process, case or decision models, and methods and styles for producing them;
- f) prescription of a particular process notation, information standard, terminology or enterprise architecture tool;
- g) description of processes whose subject is a service or organisation rather than a subject of care, such as governance, audit or quality-improvement processes.

NOTE A pathway catalogue publisher can maintain descriptions of governance processes alongside a catalogue, but they are not pathway definitions and are not covered by a conformance claim made under this document (7.1).

1.3 Audience

This section is informative.

This document is intended for those who describe, maintain, catalogue or use healthcare pathways, including clinicians, clinical informaticians, service managers, commissioners, business analysts, enterprise and data architects, integration specialists, knowledge engineers, IT system suppliers, researchers and pathway improvement teams. Chapters 5 to 16 and Appendices A, B and G are addressed to implementers of producers, consumers and catalogues. Chapter 1 and Appendices D to F are addressed to all readers.

1.4 Definition, implementation, instance and catalogue

This section is informative.

A pathway definition states the intended model of care and is the primary catalogue artefact specified by this document. A formal representation is a technology-neutral model of all or part of a definition expressed in another formalism. A pathway implementation records how an organisation or health system realises a pathway definition through local services, workforce, applications, interfaces, locations and procedures. A pathway instance is the course of care experienced by an individual subject of care. A pathway catalogue is a governed collection of pathway definitions.

Pathway instances contain personal health information and are outside the normative scope of this document; 8.29 recommends reference points that allow instance data held elsewhere to be related to pathway definitions and implementations.

1.5 Relationship with other standards

This section is informative.

This document does not replace concept systems, clinical terminologies, information models, process notations, decision languages or exchange standards. ISO 13940 provides a conceptual basis for continuity of care. HL7 FHIR can represent shareable plan definitions and implementation artefacts. BPMN, CMMN and DMN provide precise process, case and decision views, and the HL7 Field Guide to Shareable Clinical Pathways [5] gives a method and style for producing and localising such models. WHO SMART Guidelines and Digital Adaptation Kits provide structured, software-neutral guidance artefacts.

This document provides the governed, catalogue-oriented pathway business artefact. That artefact can reference formal representations expressed in those standards (8.10), be mapped to them (Appendices C to F) and be realised through implementation artefacts that use them (8.28.2). It does not depend on any of them. The mappings in Appendices C to F are informative and have not been verified by round-trip testing (Appendix N).

1.6 Conformance approach

This section is informative.

Different catalogue purposes require different degrees of formality. Four cumulative conformance levels are defined for pathway definitions: a governed catalogue record at Level 1, complete and determinate structural and behavioural relationships at Level 2, semantic binding at Level 3 and computable expressions for designated attributes under a declared expression profile at Level 4. A higher level is not inherently better. The appropriate level is determined by intended use. Obligations that apply at every level, and those that arise only at higher levels, are listed in 5.7.

1.7 Authority and intended use

This section is informative.

Conformance to this document does not establish that the clinical content of a pathway is correct or appropriate. Authority status, intended use and lifecycle state are recorded independently of one another and of technical conformance.

1.8 Conventions

1.8.1 Conformance verbs

This document uses the conformance verbs used in HL7 specifications, always in upper case:

- a) **SHALL** and **SHALL NOT** indicate a requirement that is necessary for conformance; [CONF-HPDS-001]
- b) **SHOULD** and **SHOULD NOT** indicate a recommendation; a conformant description or system can depart from it where there is a documented reason;
- c) **MAY** indicates a permission.

The words "can" and "cannot" indicate a possibility or capability and do not express conformance requirements. Conformance verbs appear only in normative text. Notes, examples, informative sections and informative appendices contain no requirements.

1.8.2 Conformance statement identifiers

Each normative statement containing **SHALL** or **SHALL NOT** is labelled with a conformance statement identifier of the form CONF-HPDS-nnn. The identifiers are assigned in document order for each working draft. They are provisional and become persistent from the first version released for formal review. Validation rules (Appendix A) have persistent identifiers of the form Vnn. Appendix M relates conformance statements, model elements and validation rules. [CONF-HPDS-002]

1.8.3 Attribute tables

In the attribute and structure tables of Chapters 7 and 8, the obligation column uses the codes M (mandatory), C (conditional) and O (optional), with the meanings given in 8.1. Multiplicity is shown as minimum..maximum, where * indicates no upper limit. The descriptions in these tables are normative.

1.8.4 Normative and informative content

Chapters 2 to 19 are normative except for notes, examples and sections marked as informative. Appendices A, B, G and J are normative. All other appendices are informative.

1.9 Implementation readiness

This section is informative.

This working draft distinguishes three things: requirements specified in the text; supporting artefacts that have actually been produced; and tests that have actually been run. Appendix I lists the examples and test cases in the WD 0.5 release package and the results obtained with the reference validator. Appendix N lists the artefacts and evidence that are still missing, including the minimum release package required for meaningful conformance testing. No statement in this document that an artefact exists, is aligned or has been validated extends beyond what is reported there.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 8601-1:2019, Date and time: Representations for information interchange: Part 1: Basic rules

ISO 13940:2026, Health informatics: System of concepts to support continuity of care

ISO/IEC 21778:2017, Information technology: The JSON data interchange syntax

IETF RFC 3986, Uniform Resource Identifier (URI): Generic Syntax

JSON Schema, Draft 2020-12, JSON Schema: A Media Type for Describing JSON Documents

NOTE ISO 13940:2026 is cited as the edition that replaces ISO 13940:2015. Its publication and final text have not been verified for this working draft (Appendix N).

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply. Terms that ISO 13940:2026 also defines, such as subject of care, are used with the meaning given in ISO 13940:2026 unless defined below.

NOTE Term 3.18 was reassigned from “task” in WD 0.4. Terms 3.57 to 3.61 are added in WD 0.5. Other term numbers are unchanged.

3.1

healthcare pathway

coordinated set of healthcare and care activities, ordered or grouped, intended to achieve defined outcomes (3.30) for a defined population (3.11) or circumstance

Note 1 to entry: A healthcare pathway can be largely sequential or largely adaptive, with activities selected according to need.

3.2

pathway definition

technology-neutral business description of the intended model of care for a healthcare pathway (3.1)

Note 1 to entry: The pathway definition is the primary business artefact (3.10) held in a pathway catalogue (3.3).

3.3

pathway catalogue

governed collection of pathway definitions (3.2) maintained for discovery, comparison, reuse, analysis and lifecycle management

3.4

catalogue entry

record within a pathway catalogue (3.3) that identifies one pathway definition (3.2) and version

3.5

pathway implementation

description of how an organisation or health system realises a pathway definition (3.2) through particular organisational and technical arrangements

3.6

pathway instance

actual course of care experienced by an individual subject of care in relation to a healthcare pathway (3.1)

3.7

pathway description

structured representation of a pathway definition (3.2) or pathway implementation (3.5) conforming to this document

3.8

reference pathway

pathway definition (3.2) published by a competent authority for reuse or specialisation (3.37)

3.9

pathway variant

pathway definition (3.2) derived from a parent pathway definition through specialisation (3.37)

3.10

business artefact

governed representation of business knowledge maintained independently of a particular technical implementation

3.11

population

set of subjects of care to whom a healthcare pathway (3.1) applies, defined by criteria

3.12

entry point

circumstance or route by which a subject of care may enter a healthcare pathway (3.1)

3.13**entry criterion**

condition required to hold for a subject of care to enter a healthcare pathway (3.1) through an entry point (3.12)

3.14**exit point**

circumstance or route by which a subject of care completes, transfers from or otherwise leaves a healthcare pathway (3.1)

3.15**exit criterion**

condition required to hold for a subject of care to leave a healthcare pathway (3.1) through an exit point (3.14)

3.16**pathway stage**

subdivision of a healthcare pathway (3.1) grouping activities (3.17), decision points (3.20) or events (3.22) at a level meaningful for business understanding

Note 1 to entry: The elements of a stage can be ordered, or available in any order while the stage is active (8.8).

3.17**activity**

action intended to be performed as part of a healthcare pathway (3.1)

Note 1 to entry: An activity does not necessarily correspond to an executable workflow task.

3.18**formal representation**

model of all or part of a pathway definition (3.2) expressed in an external process, case, decision or plan-definition formalism and maintained as a technology-neutral view of that definition

Note 1 to entry: A formal representation is distinct from an implementation artefact (3.51), which realises part of a pathway implementation (3.5).

3.19**sub-pathway**

healthcare pathway (3.1) included by reference within another healthcare pathway

3.20**decision point**

point in a healthcare pathway (3.1) at which one or more alternative pathway branches (3.21) are selected

3.21**pathway branch**

alternative route from a decision point (3.20)

3.22**event**

clinically or operationally relevant occurrence that can affect pathway behaviour

3.23**behavioural node**

pathway stage (3.16), activity (3.17), decision point (3.20) or event (3.22)

Note 1 to entry: An event whose external attribute is true has no structural container and is treated in behavioural analysis only as stated in 9.6.

3.24**service**

healthcare or care service that delivers or supports part of a healthcare pathway (3.1)

Note 1 to entry: A service is distinct from any named provider organisation.

3.25**role**

responsibility for activity, decision-making or coordination, defined independently of named individuals

3.26**care setting**

context within which care occurs

EXAMPLE Home, primary care, outpatient care, inpatient care, diagnostic facility, virtual care.

3.27

information requirement

specification of information needed to support an activity (3.17) or decision point (3.20)

3.28

information product

specification of information generated through an activity (3.17), decision point (3.20) or event (3.22)

3.29

clinical concept

unit of clinical or care meaning used within a pathway description (3.7) that can be bound to an external terminology or data specification

3.30

outcome

intended result of a healthcare pathway (3.1) or of a component of it

Note 1 to entry: An outcome is distinct from the completion of process steps.

3.31

measure

defined measurable property associated with pathway performance or achievement of an outcome (3.30)

3.32

constraint

condition restricting pathway behaviour or implementation

3.33

temporal constraint

constraint (3.32) on the timing, duration, interval, recurrence or order of pathway elements (3.47)

3.34

anchor event

event (3.22) or element (3.47) relative to which a temporal constraint (3.33) is expressed

3.35

exception

circumstance under which normal pathway behaviour does not apply, together with the alternative behaviour

3.36

evidence source

guidance, legislation, policy, research evidence, consensus statement or other source supporting pathway content

3.37

specialisation

derivation of a pathway definition (3.2) from a parent pathway definition while retaining inherited content and recording deviations (3.38)

3.38

deviation

recorded difference between a pathway variant (3.9) or pathway implementation (3.5) and the pathway definition (3.2) from which it derives

3.39

terminology binding

association between an element (3.47) and an identified, versioned concept, value set or data element in an external semantic resource

3.40

local term

concept used within a pathway description (3.7) that is not bound to an external semantic resource and is identified as local

3.41**interoperability requirement**

technology-neutral specification of an information exchange on which a healthcare pathway (3.1) depends

3.42**authority status**

declared level of professional or organisational authority applying to a pathway description (3.7)

3.43**intended use**

declared purpose for which a pathway description (3.7) is intended to be relied upon

3.44**clinical authority**

body or individual with professional authority to approve the clinical content of a pathway description (3.7)

3.45**pathway owner**

organisation or role accountable for a pathway description (3.7)

3.46**pathway steward**

organisation or role responsible for maintaining a pathway description (3.7)

3.47**element**

instance of any entity of the conceptual model within a pathway description (3.7)

3.48**structural relationship**

relationship between elements (3.47) expressing containment or inclusion

3.49**behavioural relationship**

relationship between elements (3.47) expressing order, concurrency, choice, repetition, waiting, interruption, escalation or transition

3.50**provenance record**

record of the origin, authorship, approval and derivation of a pathway description (3.7) or of an element (3.47) within it

3.51**implementation artefact**

workflow definition, case model, decision model, decision support rule, configuration or other technical artefact that realises part of a pathway implementation (3.5)

3.52**domain profile**

set of additional constraints, required bindings and extensions (3.54) applied to this document for a defined domain or use case

3.53**expression profile**

profile defining the expression language, evaluation context and semantics required for interoperable computable expressions

3.54**extension**

identified addition to the conceptual model

Note 1 to entry: A modifying extension changes the interpretation of the element (3.47) that carries it; a non-modifying extension does not.

3.55**pathway description producer**

software or process that creates or modifies pathway descriptions (3.7)

3.56**pathway description consumer**

software or process that reads or interprets pathway descriptions (3.7)

3.57**governance event**

dated record of a change to the governance state of a published pathway description (3.7), such as a lifecycle transition, a change of authority status, an approval or a review outcome

Note 1 to entry: Governance events are appended; they do not change the content of the version (14.2).

3.58**governance-state attribute**

attribute of a pathway description (3.7) whose value can change after publication, but only by a governance event (3.57)

3.59**behavioural graph**

directed graph formed by the behavioural nodes (3.23), pathway branches (3.21), entry points (3.12) and exit points (3.14) of a pathway definition (3.2) and the behavioural relationships (3.49) between them, together with the containment of behavioural nodes in pathway stages (3.16)

3.60**initial node**

behavioural node (3.23) directly contained in an ordered pathway stage (3.16) that has no incoming behavioural relationship (3.49) from another element directly contained in the same stage

Note 1 to entry: Initial nodes commence when their stage commences (9.6).

3.61**representation discrepancy**

recorded difference between a formal representation (3.18) and the pathway definition (3.2) it represents that contradicts the definition or whose classification has not been determined

4 Abbreviated terms

API	application programming interface
BPMN	Business Process Model and Notation
CMMN	Case Management Model and Notation
CPG	Clinical Practice Guidelines (HL7 FHIR implementation guide)
CQL	Clinical Quality Language
DAK	Digital Adaptation Kit
DMN	Decision Model and Notation
EPR	electronic patient record
FHIR	Fast Healthcare Interoperability Resources
HL7	Health Level Seven
HPDS	Healthcare Pathway Description Standard
ICD	International Classification of Diseases
JSON	JavaScript Object Notation
JSON-LD	JSON for Linking Data
URI	Uniform Resource Identifier

5 Conformance

5.1 Conformance classes

Conformance **MAY** be claimed by a pathway definition, a pathway implementation, a pathway catalogue, a pathway description producer or a pathway description consumer. A claim **SHALL** identify the version of this document, the conformance class, the conformance level where the class is pathway definition, and any domain or expression profiles applied, with their versions. A claim of Level 3 or Level 4 **SHALL** also include the binding report specified in 11.5. A claim **SHOULD** identify the release package, if any, with which the claimed description was validated. [CONF-HPDS-003]

A conformance claim concerns the representation of a pathway only. It **SHALL NOT** be presented as, or used to imply, clinical approval, authority or fitness for a particular use, which are recorded separately (Chapter 16). [CONF-HPDS-004]

5.2 Pathway definition conformance levels

A pathway definition **SHALL** declare exactly one conformance level from Table 1. Each level includes all requirements of the levels below it. [CONF-HPDS-005]

Table 1: Pathway definition conformance levels

Level	Name	Requirements
1	Catalogue-ready	Contains the minimum pathway definition dataset (5.3), persistent identity, governance metadata and a human-readable catalogue record (Chapter 17). Principal stages, activities, decision points, roles and outcomes MAY be represented without a complete behavioural graph. Relationships that are present are of permitted types (V17), and relationships to other pathways MAY be recorded at pathway level (9.2). The graph MAY be incomplete (9.9). All Level 1 validation rules in Appendix A are met.
2	Structured	All represented stages, activities, decision points, branches, events, entry and exit points, roles, services, care settings, information requirements, information products and outcomes are distinct identified elements. Structural and behavioural relationships are complete and determinate as specified in 9.3 to 9.9, using ordered or unordered stages as appropriate. All Level 2 validation rules are met.
3	Semantically bound	The clinical concepts on which population, entry and exit criteria and mandatory information requirements depend are identified (8.26.2, 11.3) and carry versioned terminology or data-specification bindings, or are declared local terms. A binding report distinguishes bound from local concepts (11.5). All Level 3 validation rules are met.
4	Computably specified	Every expression-bearing attribute designated in Table 52 carries a computable expression conforming to a declared expression profile (13.4), subject to the professional-judgement and authorised-exception provisions of 13.7. Every decision point that is not judgement-based carries logic provenance. All Level 4 validation rules are met.

NOTE 1 A higher level does not indicate a better pathway definition. A definition used for catalogue discovery or service design can be fit for purpose at Level 1 or Level 2.

NOTE 2 Level 4 conformance does not imply that the pathway, or any part of it, is executable. It establishes only that the designated expressions are computable under a declared profile. Execution and orchestration semantics are outside the scope of this document (1.2, 9.7, 16.3). A definition at any level can reference formal representations (8.10); doing so does not change its conformance level.

5.3 Minimum pathway definition dataset

Every pathway definition **SHALL** contain the items in Table 2, irrespective of conformance level. [CONF-HPDS-006]

Table 2: Minimum pathway definition dataset

Item	Represented by	Reference
Pathway identifier	HealthcarePathway.identifier	8.3
Pathway name	HealthcarePathway.name	8.3
Version	HealthcarePathway.version	8.4, Chapter 15
Lifecycle status	HealthcarePathway.lifecycleStatus	8.4, 14.5
Authority status	HealthcarePathway.authorityStatus	8.4, 16.1
Intended use	HealthcarePathway.intendedUse	8.4, 16.2
Purpose	HealthcarePathway.purpose	8.4

Item	Represented by	Reference
Owner	HealthcarePathway.owner	8.4
Target population	Population	8.5
Entry points and entry criteria	EntryPoint	8.6
Exit points and exit criteria	ExitPoint	8.7
Principal activities	Activity	8.9
Principal decision points, or a declaration that none exist	DecisionPoint or HealthcarePathway.noDecisionPoints	8.4, 8.11
Responsible roles	Role	8.14
Expected outcomes	Outcome	8.20
Applicable evidence or guidance	EvidenceSource	8.24

A pathway definition **SHOULD** also identify its principal stages (8.8) where the pathway has a recognisable stage structure.

NOTE Principal stages were mandatory in WD 0.3. Testing against a catalogue of adaptive, cross-organisational pathways showed that many such pathways have no stable stage structure at catalogue level and that imposing one produced artificial content (K.5).

5.4 Pathway implementation conformance

A conformant pathway implementation **SHALL** reference one pathway definition and its version, identify its implementing organisation or organisations, map relevant definition elements to local implementation artefacts, record every deviation from the realised definition explicitly, and maintain governance and versioning metadata independent of the definition. A pathway implementation **SHALL NOT** alter the pathway definition it realises. All validation rules designated for implementations in Appendix A **SHALL** be met. [CONF-HPDS-007]

5.5 Pathway catalogue conformance

A conformant pathway catalogue **SHALL** meet the following requirements: [CONF-HPDS-008]

- each catalogue entry **SHALL** reference a conformant pathway definition by identifier and version; [CONF-HPDS-009]
- catalogue entry identifiers **SHALL** be unique within the catalogue and **SHALL** remain stable across catalogue releases; [CONF-HPDS-010]
- catalogue metadata **SHALL** identify publisher, governance authority, scope and catalogue version; [CONF-HPDS-011]
- catalogue search and export **SHALL** preserve pathway identifiers and versions; [CONF-HPDS-012]
- a catalogue **SHALL NOT** represent implementation-specific content as part of a pathway definition; [CONF-HPDS-013]
- all validation rules designated for catalogues in Appendix A **SHALL** be met. [CONF-HPDS-014]

5.6 Producer and consumer conformance

A conformant pathway description producer **SHALL**: [CONF-HPDS-015]

- produce descriptions that conform to the level claimed for them;
- assign identifiers in accordance with Chapter 15;
- preserve without alteration all extensions and elements it does not interpret when modifying an existing description;
- apply the validation rules of Appendix A before publication and report errors and warnings;
- record a change record (14.4) for every published change.

A conformant pathway description consumer **SHALL**: [CONF-HPDS-016]

- accept any description that conforms at Level 1 and is serialised in accordance with Chapter 19;
- preserve identity and version metadata when storing, displaying or exporting descriptions;
- ignore, without error, non-modifying extensions it does not support;

- d) not interpret content affected by a modifying extension it does not support for any purpose that depends on the meaning of that content, at any conformance level; it **MAY** store, preserve and display such content only as permitted by 13.6;
- e) display authority status and intended use whenever a description is presented;
- f) not imply clinical authority beyond that recorded in the description;
- g) handle differences between a formal representation and the pathway definition it represents as specified in 8.10.1 to 8.10.3, and not rely on a formal representation for an element affected by an open representation discrepancy;
- h) when presenting a description claimed at Level 3 or Level 4, distinguish concepts bound to external semantic resources from declared local terms (11.5);
- i) determine the current and historical governance state of a published description as specified in 14.6.

5.7 Obligations by conformance level

Table 3 states which obligations apply at every conformance level and which arise only at higher levels. An obligation stated elsewhere in this document without a level qualification applies at every level, except where Table 3, Table 47 or Appendix A states otherwise.

Table 3: Obligations by conformance level

Obligation	Applies from	Validation rules
Minimum pathway definition dataset, identification, governance metadata, authority status and intended use	Level 1	V01, V02, V03, V04 to V08, V10 to V15
Attribute obligations and multiplicities for every element present	Level 1	V59
No implementation-specific delivery arrangements in the definition or its formal representations	Level 1	V09, V54
Relationships present are of permitted types and within maximum cardinalities	Level 1	V17
Cross-description targets version qualified	Level 1	V32
Formal representations identified; discrepancies recorded	Level 1	V53, V66
Unordered stages have completion conditions	Level 1	V56
Governance events consistent with governance-state attributes; publication event present	Level 1	V62, V63
Any computable expression present identifies its profile, language and version	Level 1	V42
Every represented item is a distinct identified element; minimum relationship cardinalities met	Level 2	V16, V17 (Table 47 NOTE 2)
Containment, entry, decision and branch structure, join rules, responsibility and information requirements	Level 2	V18, V20 to V27, V31, V33
Reachability, termination and cycles in the behavioural graph	Level 2	V28 to V30, V64, V65, V67, V68
Relationships to other pathways refined to exit points, events and activities	Level 2	V55, V60
Deviations of a variant complete	Level 2	V37, V61
Clinical concepts of criteria identified and bound or declared local; binding report	Level 3	V38 to V41; 11.5
Designated expressions computable; logic provenance; expression profile content	Level 4	V43 to V45

A simple catalogue record at Level 1 does not require a behavioural graph, containment relationships, responsibility relationships or information requirements.

6 Conceptual framework

6.1 Separation of definition, implementation and instance

Whether a pathway description describes a pathway definition or a pathway implementation is declared by its root element: HealthcarePathway for a definition and PathwayImplementation for an implementation (Appendix B). A pathway description **SHALL NOT** contain personal health information. [CONF-HPDS-017]

A pathway definition **SHALL** describe the intended model of care without implementation-specific delivery arrangements. Implementation-specific delivery arrangements are: named provider organisations or teams identified as delivering services, activities or decisions; named locations or sites; applications, interfaces and devices identified by product, supplier or instance; worklists and order sets; local procedures; and named individuals in performing or deciding roles. Such information **SHALL** be recorded in a PathwayImplementation (8.27). [CONF-HPDS-018]

The restriction applies to the care-model content of a definition: its Population, entry and exit, stage, activity, decision, branch, event, role, service, care setting, information, outcome, measure, constraint, exception and interoperability elements and their relationships. It does not apply to governance and provenance metadata, in which organisations and people **MAY** be named. These are: owner, steward, clinicalAuthority, approvals, rights, governance events, ProvenanceRecord and ChangeRecord authors, EvidenceSource citations, and the publisher and governanceAuthority of a PathwayCatalogue.

NOTE A definition does not fail conformance because its owner, steward, approver or clinical authority is a named organisation.

The same separation applies to formal representations. A formal representation referenced by a pathway definition (8.10) **SHALL** contain no implementation-specific delivery arrangements, in the sense defined above, although its own governance and provenance metadata **MAY** name organisations and people. A model that has been localised to particular organisations, systems or procedures is an implementation artefact (8.28.2) and **SHALL** be recorded against a pathway implementation. [CONF-HPDS-019]

6.2 Design principles

The requirements of this document are derived from the principles in Table 4. The principles are informative, but they **SHOULD** be used to resolve questions of interpretation that the normative text does not settle.

Table 4: Design principles

Principle	Statement
Catalogue first	The primary unit is a governed pathway definition that can be discovered, compared, versioned and reused in a catalogue.
Technology neutrality	The pathway definition is independent of workflow engine, EPR, modelling tool, supplier platform and exchange technology. Local organisational and technology choices are linked through implementation descriptions rather than embedded in the definition.
Business meaning first	Pathways are described in terms recognisable to clinicians, managers and architects responsible for care and service design.
Human and machine readability	The same underlying description supports understandable catalogue views and structured processing.
Minimum sufficient structure	Entities, attributes and requirements are included only where they have demonstrable value for describing, comparing or governing pathways.
Explicit relationships	Containment, inclusion, behaviour, responsibility, information dependency, evidence and outcomes are represented explicitly, so that activities and decisions can be traced to evidence, roles, information, implementations and measures.
Semantic openness	Important concepts can be bound to recognised terminologies and information standards without requiring one universal terminology or information model.
Modularity and reuse	Stages, activities, decisions and sub-pathways are reusable and are referenced rather than duplicated.
Variation without fragmentation	Specialised pathways retain parent identity and record deviations rather than becoming unrelated copies.
Lifecycle governance	Published pathway definitions are immutable versions with traceable identity, origin, ownership, approval, authority and change history.
Progressive formality	Greater computability is available where it is needed but is not inherently better. Formal representations add precision without becoming the source of pathway meaning.

6.3 Level of detail

The depth of decomposition **SHOULD** be determined by the business purpose of the catalogue. Activities are not decomposed into tasks; where finer structure is needed, an activity **SHOULD** be represented as a stage containing activities, or as an included sub-pathway. A pathway catalogue **SHOULD** remain usable without exposing the internal configuration detail of execution systems.

7 Catalogue model

7.1 PathwayCatalogue

A PathwayCatalogue is a governed collection of pathway definitions. A catalogue manifest **SHALL** reference its entries by identifier and version and **SHALL NOT** embed or duplicate their content. This allows a pathway definition to be listed in several catalogues while retaining one stable identity. A PathwayCatalogue carries the common attributes in 8.3 and the attributes in Table 5. [\[CONF-HPDS-020\]](#)

Table 5: PathwayCatalogue attributes

Attribute	Obl.	Mult.	Type	Description
version	M	1..1	Text	Version of the catalogue manifest.
publisher	M	1..1	Text	Organisation responsible for publishing the catalogue.
scope	M	1..1	Text	Scope of the catalogue.
governanceAuthority	M	1..1	Text	Authority responsible for catalogue governance.
entry	M	1..*	CatalogueEntry	Catalogue entries (7.2).
classificationScheme	O	0..*	Reference	Classification schemes or taxonomies used for discovery.
profile	O	0..*	Reference	Domain or expression profiles required of entries, with versions.
releaseDate	O	0..1	Date	Date of this catalogue release.
rights	O	0..1	Text	Licence, copyright and conditions of reuse applying to the catalogue manifest.

NOTE Catalogue entries reference pathway definitions only (5.5). Where a publisher also maintains descriptions of governance or improvement processes whose subject is a service rather than a subject of care, those descriptions are held outside the catalogue manifest or clearly distinguished from it, and are not covered by the catalogue conformance claim.

7.2 CatalogueEntry

A CatalogueEntry identifies one pathway definition version within a catalogue. CatalogueEntry is a structured data type (8.1): it carries the entryIdentifier component but not the common attributes. Its structure is given in Table 6.

Table 6: CatalogueEntry structure

Component	Obl.	Mult.	Type	Description
entryIdentifier	M	1..1	Identifier	Identifier of the entry, unique within the catalogue and stable across releases.
pathway	M	1..1	Reference	Pathway definition referenced, including its version.
entryStatus	M	1..1	Enumeration	current or withdrawn. A withdrawn entry remains listed for historical interpretation.
addedDate	O	0..1	Date	Date the entry was added to the catalogue.
classification	O	0..*	CodeableText	Catalogue-specific classification of the entry.

7.3 Catalogue discovery

A conformant catalogue **SHOULD** support discovery using pathway name, identifier, pathway type, population, owner, service, care setting, intended use, authority status, lifecycle status, outcome and applied profile. Discovery metadata **SHALL** be derived from the referenced pathway definitions and **SHALL NOT** become an independent source of pathway meaning. [\[CONF-HPDS-021\]](#)

8 Pathway conceptual model

8.1 General

The entity types and structured data types listed in Table 7, the Relationship entity and relationship types in Chapter 9, and the data types in 8.2 form the normative conceptual model. The conceptual model is authoritative over serialisations, schemas and mappings (19.3). Differences between a definition and a formal representation are handled as specified in 8.10.

An element is an instance of an entity type. Every element carries the common attributes (8.3) and can be the target of a Reference. A structured data type is a value held within an element; it does not carry the common attributes. A structured data type that has an identifier component (CatalogueEntry, FormalRepresentation, Deviation, ImplementationArtefact and RepresentationDiscrepancy) can also be the target of a Reference.

Table 7: Entity types and structured data types

Category	Types
Entity types (carry common attributes)	PathwayCatalogue; HealthcarePathway; Population; EntryPoint; ExitPoint; PathwayStage; Activity; DecisionPoint; PathwayBranch; Event; Role; Service; CareSetting; InformationRequirement; InformationProduct; ClinicalConcept; Outcome; Measure; Constraint; Exception; EvidenceSource; InteroperabilityRequirement; PathwayImplementation; Relationship
Structured data types with an identifier component	CatalogueEntry; FormalRepresentation; Deviation; ImplementationArtefact; RepresentationDiscrepancy
Other structured data types	TerminologyBinding; Expression; TemporalConstraint; RecurrenceSpec; IterationSpec; ProvenanceRecord; ApprovalRecord; ChangeRecord; Extension; GovernanceEvent; ImplementationMapping

A pathway definition consists of exactly one HealthcarePathway element and collections of elements of the other entity types that belong to a definition, including Relationship elements. A pathway implementation consists of exactly one PathwayImplementation element and any local elements and relationships it needs. The collections and their serialised names are given in Appendix B. A collection **MAY** be empty unless a requirement states otherwise.

In the attribute tables, the obligation column uses the codes M (mandatory), C (conditional) and O (optional). An attribute marked C is mandatory when the condition stated in its description holds and optional otherwise; where the condition holds, its minimum multiplicity is at least 1. Multiplicity is shown as minimum..maximum, where * indicates no upper limit. Where an optional Boolean attribute is absent, its value is false. Attribute obligations and multiplicities are validated by V59.

8.2 Data types

Table 8: Data types

Data type	Definition
Identifier	Globally unique or namespace-unique persistent identifier, preferably a URI conforming to IETF RFC 3986.
Text	Human-readable character string, with optional language tag.
CodeableText	Human-readable text with zero or more terminology bindings.
Date	Calendar date conforming to ISO 8601-1:2019.
Duration	Duration conforming to ISO 8601-1:2019.
Reference	Identifier of another element, qualified by the version, or permitted version range, of the pathway description in which the element is defined where that description differs from the referencing description. In the normative serialisation its components are ref, pathwayDescription, version and versionRange. A Reference held by one of the following attributes identifies a resource that is not an element of a pathway description, by its identifier in ref and its version in version, and has no pathwayDescription component: HealthcarePathway.profile, PathwayCatalogue.profile, PathwayCatalogue.classificationScheme, Expression.expressionProfile, Activity.applicableStandard, Activity.definitionReference, InformationRequirement.informationStandard, InformationProduct.informationStandard, InteroperabilityRequirement.informationStandard, ClinicalConcept.valueSet, Measure.measureReference and ImplementationArtefact.standard. Such references are not resolved against pathway descriptions (V16).
Enumeration	Value drawn from a value set specified in this document. A domain profile MAY restrict a value set but SHALL NOT extend it (13.3).
Boolean	True or false.

Data type	Definition
Integer	Non-negative whole number.
Structured data types	FormalRepresentation (8.10); TerminologyBinding, Expression, TemporalConstraint, RecurrenceSpec, IterationSpec, ProvenanceRecord, ApprovalRecord, ChangeRecord, Deviation, Extension, GovernanceEvent and RepresentationDiscrepancy (8.26); CatalogueEntry (7.2); ImplementationMapping and ImplementationArtefact (8.28). See Table 7.

8.3 Common attributes

Every element of an entity type listed in Table 7, including every Relationship (9.1), **SHALL** carry the common attributes in Table 9 in addition to the attributes specified for it. Structured data types do not carry the common attributes. [CONF-HPDS-022]

Table 9: Common attributes

Attribute	Obl.	Mult.	Type	Description
identifier	M	1..1	Identifier	Persistent identifier of the element (Chapter 15).
name	C	0..1	Text	Human-readable display name. Mandatory for every element other than a Relationship. A change of name does not change the identifier.
description	O	0..1	Text	Narrative description.
classification	O	0..*	CodeableText	Catalogue classification or subject tags.
conceptBinding	O	0..*	TerminologyBinding	Terminology or information-model bindings applying to the element (Chapter 11).
localTerm	O	0..1	Boolean	True where the principal meaning is intentionally represented by a local term.
provenance	C	0..*	ProvenanceRecord	Element-level provenance. Mandatory where the origin of the element differs materially from that of the description as a whole (14.3).
evidence	O	0..*	Reference	EvidenceSource elements supporting the element.
extension	O	0..*	Extension	Declared extensions (13.6).

8.4 HealthcarePathway

HealthcarePathway is the root business artefact of a pathway definition and the principal unit referenced by a pathway catalogue. A pathway definition **SHALL** contain exactly one HealthcarePathway element. [CONF-HPDS-023]

Table 10: HealthcarePathway attributes

Attribute	Obl.	Mult.	Type	Description
version	M	1..1	Text	Business version of this pathway definition (Chapter 15).
lifecycleStatus	M	1..1	Enumeration	Current lifecycle state (14.5). Governance-state attribute (14.2).
authorityStatus	M	1..1	Enumeration	Current authority status of the content (16.1). Governance-state attribute (14.2).
intendedUse	M	1..*	Enumeration	Purposes for which the definition is intended to be relied upon (16.2).
conformanceLevel	M	1..1	Integer	Declared conformance level, 1 to 4.
purpose	M	1..1	Text	What the pathway is intended to achieve.
scope	O	0..1	Text	What the pathway covers and excludes.
pathwayType	O	0..*	CodeableText	Classification of the pathway (see NOTE 1).
owner	M	1..1	Text	Organisation or role accountable for the pathway definition.
steward	O	0..1	Text	Organisation or role responsible for maintaining the definition.
clinicalAuthority	O	0..*	Text	Clinical or professional authority approving the clinical content, as recorded when the content was fixed. The requirements of Table 54 for the current authority status are evaluated over this attribute and the governance events (16.1, V06).
approval	O	0..*	ApprovalRecord	Approvals recorded when the content was fixed. Approvals given after publication are recorded in governance events. The requirements of Table 54 are evaluated over both (16.1, V06).
effectiveDate	C	0..1	Date	Date from which the definition applies. Mandatory where lifecycleStatus is published, under-review, superseded or retired.

Attribute	Obl.	Mult.	Type	Description
reviewDate	C	0..1	Date	Current date by which the definition is due for review. Mandatory under the same condition as effectiveDate. Governance-state attribute (14.2).
publicationDate	O	0..1	Date	Date of publication.
rights	O	0..1	Text	Licence, copyright and conditions of reuse and derivation. Should be present where the definition is published for use outside the owning organisation.
parent	C	0..1	Reference	Parent pathway definition, with version. Mandatory where the definition is a pathway variant (Chapter 13).
specialisationLevel	O	0..1	Enumeration	international, national, regional, organisational, service or other.
profile	O	0..*	Reference	Domain or expression profiles applied, with versions. At least one expression profile is mandatory where conformanceLevel is 4 (13.4).
representation	O	0..*	FormalRepresentation	Formal representations of all or part of the definition (8.10).
supersedes	O	0..*	Reference	Earlier pathway definitions or versions superseded by this version.
noDecisionPoints	C	0..1	Boolean	True where the pathway contains no decision points. Mandatory where no DecisionPoint element is present.
modifyingExtension	C	0..*	Identifier	Identifiers of modifying extensions used anywhere in the definition. Mandatory where any modifying extension is used (13.6).
changeHistory	C	0..*	ChangeRecord	History of published changes. Mandatory for every version after the first published version (14.4).
deviation	C	0..*	Deviation	Recorded differences from the parent definition. Mandatory where parent is present, with the entries required by 13.2.
governanceEvent	C	0..*	GovernanceEvent	Governance events recorded for this version, in non-decreasing date order. At least one event of type publication is mandatory where lifecycleStatus is published or any later state (14.2). Governance-state attribute.

NOTE 1 An informative value set for pathwayType includes: diagnostic, procedural, treatment, long-term condition management, care coordination, screening, prevention, maternity, medicines, rehabilitation, mental health, urgent and emergency care, elective care and end-of-life care. Domain profiles may constrain this value set.

NOTE 2 Owner, steward and clinical authority are recorded separately because organisational accountability, maintenance responsibility and approval of clinical content often rest with different parties. Accountability for coordinating care within the pathway is a different matter again and is recorded with an accountableFor relationship from a Role (9.2).

8.5 Population

A pathway definition **SHALL** contain at least one Population element. Population defines the set of subjects of care to whom the pathway applies. [\[CONF-HPDS-024\]](#)

Table 11: Population attributes

Attribute	Obl.	Mult.	Type	Description
inclusionCriterion	M	1..*	Expression	Criteria for inclusion in the population, including clinical, demographic, risk and eligibility criteria.
exclusionCriterion	O	0..*	Expression	Criteria excluding a subject of care from the population.
stratification	O	0..*	Expression	Risk, need or other stratification that varies the intended care within the population.

NOTE WD 0.3 provided separate attributes for demographic, clinical-state, risk and eligibility criteria. They are expressed as inclusion or exclusion criteria and can be distinguished, where needed, by terminology binding or domain profile (K.5).

8.6 EntryPoint

A pathway definition **SHALL** contain at least one EntryPoint element. The entry point records the route of arrival and the entry criteria record the conditions that **SHALL** hold; the two **SHALL NOT** be combined in one attribute. The behavioural node at which the pathway starts for an entry point is recorded by a startsAt relationship (9.2). [\[CONF-HPDS-025\]](#)

Table 12: EntryPoint attributes

Attribute	Obl.	Mult.	Type	Description
entryType	M	1..1	Enumeration	referral, self-referral, diagnosis, presentation, screening-result, case-finding, scheduled-review, transition or other.
criterion	M	1..*	Expression	Entry criteria applying at this entry point.
source	O	0..*	CodeableText	Originating service, care setting or pathway.
fromPathway	C	0..1	Reference	Originating pathway, with version. Mandatory where entryType is transition.

8.7 ExitPoint

A pathway definition **SHALL** contain at least one ExitPoint element. A pathway definition **SHOULD** represent every exit type in Table 13 that can apply to the pathway, not only the intended outcome. Transfer into another pathway is recorded by a transitionsTo relationship (9.2). [CONF-HPDS-026]

Table 13: ExitPoint attributes

Attribute	Obl.	Mult.	Type	Description
exitType	M	1..1	Enumeration	outcome-achieved, transfer, discharge, escalation, patient-choice, clinical-exception, death or other.
criterion	M	1..*	Expression	Exit criteria applying at this exit point.
outcome	O	0..*	Reference	Outcomes associated with this exit.
informationProduct	O	0..*	Reference	Information products generated at exit.

NOTE Continuing-care and coordination pathways often have no single completion point. Their exits are typically transfer, patient-choice, death, or cessation of benefit recorded as outcome-achieved or other, and are to be stated explicitly rather than omitted.

8.8 PathwayStage

A PathwayStage is a clinically or operationally meaningful subdivision used to organise pathway content. Stages **MAY** be nested. Care settings are related to stages by occursIn relationships.

The ordering attribute states how the behavioural nodes directly contained in the stage become available when the stage commences. In an ordered stage, behaviour among the contained nodes is expressed by behavioural relationships; the initial nodes of the stage (3.60) commence when the stage commences (9.6). In an unordered stage, every directly contained node is available when the stage commences, subject to its own prerequisites, entry conditions and triggering events; precedes relationships between contained nodes **MAY** be used where a specific ordering constraint applies. A nested stage follows its own ordering once it is available.

A stage is complete when its completion conditions hold. Where an ordered stage has no completion condition, it is complete when each of its terminal nodes that is not an optional activity has completed or has not been selected by any decision on the route taken; a terminal node is a directly contained node with no outgoing behavioural relationship to another element directly contained in the same stage. An unordered stage **SHALL** state at least one completion condition. Optional activities do not affect whether a stage is complete. These rules describe intended behaviour for analysis and presentation; they do not define run-time behaviour (9.8). [CONF-HPDS-027]

Table 14: PathwayStage attributes

Attribute	Obl.	Mult.	Type	Description
ordering	O	0..1	Enumeration	ordered or unordered. Absence means ordered.
sequenceLabel	O	0..1	Text	Presentation order label only. Behaviour SHALL NOT be inferred from it.
purpose	O	0..1	Text	Purpose of the stage.
entryCondition	O	0..*	Expression	Conditions for commencing the stage.
completionCondition	C	0..*	Expression	Conditions under which the stage is complete. Mandatory where ordering is unordered.

NOTE An unordered stage describes adaptive, need-driven work in business terms. It corresponds broadly to a CMMN stage or a BPMN ad hoc sub-process (Appendix D), but this document does not adopt the execution semantics of either.

8.9 Activity

An Activity is an action intended to be performed as part of the pathway. An activity is not decomposed into tasks. Where finer structure is needed, the activity **SHOULD** be represented as a PathwayStage containing activities, or as an included sub-pathway (6.3). An activity realised by a sub-pathway is the source of an includes relationship to that sub-pathway (9.3).

Table 15: Activity attributes

Attribute	Obl.	Mult.	Type	Description
purpose	O	0..1	Text	Purpose of the activity.
activityType	O	0..1	CodeableText	Kind of activity, such as assessment, investigation, procedure, referral, prescribing, education, coordination or review.
requiresClinicalResponsibility	M	1..1	Boolean	Whether the activity requires an identified clinically responsible role. Where true, at least one performs relationship is required (V25).
optional	O	0..1	Boolean	True where the activity is optional or discretionary.
prerequisite	O	0..*	Expression	Conditions that SHALL hold before the activity is performed.
expectedDuration	O	0..1	Duration	Expected duration.
completionCriterion	O	0..*	Expression	Conditions under which the activity is complete.
applicableStandard	O	0..*	Reference	Applicable professional, information or technical standards.
definitionReference	O	0..1	Reference	Reusable activity definition from which this activity derives.

8.10 FormalRepresentation

A FormalRepresentation identifies a model of all or part of a pathway definition expressed in an external formalism, such as a BPMN process, a CMMN case, a DMN decision model, a FHIR PlanDefinition, a CQL library or WHO DAK content. It allows a pathway definition to carry greater behavioural or decision precision without this document defining that precision itself.

The following requirements apply:

- a formal representation **SHALL** be technology neutral in the sense of 6.1; a model localised to particular organisations, locations, applications, interfaces or local procedures **SHALL** instead be recorded as an ImplementationArtefact of a pathway implementation (8.28.2); [\[CONF-HPDS-028\]](#)
- a formal representation **SHALL NOT** be the only place in which an element or relationship required for the declared conformance level is recorded; [\[CONF-HPDS-029\]](#)
- differences between a formal representation and the pathway definition **SHALL** be classified and handled as specified in 8.10.1 to 8.10.3; [\[CONF-HPDS-030\]](#)
- a formal representation acquires no clinical authority beyond that recorded for the pathway definition, and referencing one does not make the pathway executable (9.7).

Table 16: FormalRepresentation structure

Component	Obl.	Mult.	Type	Description
identifier	M	1..1	Identifier	Identifier of the representation.
representationType	M	1..1	Enumeration	process-model, case-model, decision-model, plan-definition, logic-library, guideline-package or other.
standard	M	1..1	Text	Standard and version in which the representation is expressed, such as BPMN 2.0.2, CMMN 1.1, DMN 1.5 or HL7 FHIR R5 PlanDefinition.
locator	M	1..1	Text	URI or other locator of the representation.
representationVersion	M	1..1	Text	Version of the representation.
representsElement	O	0..*	Reference	Elements of the definition represented. Absence means the representation covers the pathway as a whole.

Component	Obl.	Mult.	Type	Description
correspondence	M	1..1	Enumeration	derived (produced from the definition), source (the definition was produced from it) or aligned (maintained independently and reconciled).
method	O	0..1	Text	Modelling method or style applied, such as HL7 Field Guide to Shareable Clinical Pathways, Edition 3.
note	O	0..1	Text	Explanation of permitted elaboration in the representation and of its coverage (8.10.1).
discrepancy	C	0..*	RepresentationDiscrepancy	Contradictions and unresolved discrepancies between the representation and the definition (8.10.1). Mandatory for each such difference known to the producer. Governance-state attribute (14.2).

NOTE A shareable clinical pathway produced in accordance with the HL7 Field Guide [5] is a platform-independent set of BPMN, CMMN and DMN models. Before localisation it is a formal representation of a pathway definition; once localised for a particular organisation it is an implementation artefact (Appendix D).

8.10.1 Classes of difference

A difference between a formal representation and the pathway definition it represents is one of the following:

- contradiction: the representation asserts meaning that is incompatible with the definition for an element it represents, such as a different population, criterion, decision outcome, ordering, responsibility or intended outcome;
- elaboration: the representation adds detail that this document does not represent, such as decomposition of an activity, user-interface steps or data mappings, without contradicting the definition;
- incomplete coverage: the representation does not represent some elements of the definition, as declared by representsElement;
- unresolved discrepancy: a difference whose classification as contradiction or elaboration has not yet been determined.

Elaboration and incomplete coverage are permitted and are not defects; they **SHOULD** be explained in the note component. Contradictions and unresolved discrepancies **SHALL** be recorded as RepresentationDiscrepancy entries (8.26.11). The correspondence component records only the declared relationship of derivation (derived, source or aligned); it is not used to record differences. [\[CONF-HPDS-031\]](#)

8.10.2 Resolving discrepancies

The approved meaning of a pathway is that of the pathway definition, subject to its authority status and intended use, whatever the declared correspondence. A contradiction is resolved by issuing a new version of the definition, a new version of the representation, or both, and recording the resolution in the discrepancy entry.

Where correspondence is source, a contradiction indicates an error in the derivation of the definition or a later change to the representation. It **SHALL NOT** be resolved by treating the representation as authoritative; the owner of the definition **SHALL** decide which meaning is intended and issue a new version accordingly. Where correspondence is derived, a contradiction is presumed to be a defect in the representation. Where correspondence is aligned, neither is presumed defective until the discrepancy is resolved. [\[CONF-HPDS-032\]](#)

8.10.3 Reliance while a discrepancy is open

While a discrepancy of type contradiction or unresolved has status open, for each affected element:

- a consumer **MAY** rely on the pathway definition, to the extent permitted by its authority status and intended use;
- a consumer **SHALL NOT** rely on the formal representation; [\[CONF-HPDS-033\]](#)
- a presentation of the formal representation, or of the affected element, **SHALL** indicate that an open discrepancy exists. [\[CONF-HPDS-034\]](#)

A consumer **MAY** rely on elaborations that do not contradict the definition, to the extent permitted by the governance of the representation. Open discrepancies are reported by V66.

8.11 DecisionPoint

A DecisionPoint is a point at which one or more alternative pathway branches are selected. Responsible roles are recorded by decides relationships. Where selection relies on professional judgement that is not intended to be reduced to a rule, the decision criteria **SHALL** be recorded as narrative and judgementBased **SHALL** be true. The judgementBased attribute applies to the decision criteria of the decision point and to the conditions of every branch it selects, and to nothing else (13.7). [\[CONF-HPDS-035\]](#)

The branches of a decision point define its permitted outcomes. A pathway variant or pathway implementation that adds, removes or substitutes an outcome **SHALL** record the change as a Deviation (13.2). [\[CONF-HPDS-036\]](#)

Table 17: DecisionPoint attributes

Attribute	Obl.	Mult.	Type	Description
decisionQuestion	M	1..1	Text	Question to be decided.
decisionCriterion	M	1..*	Expression	Criteria for selection among alternatives.
selectionRule	M	1..1	Enumeration	exclusive (exactly one branch), inclusive (one or more branches) or deferred (the branch whose condition refers to an event is selected when that event occurs).
judgementBased	O	0..1	Boolean	True where selection relies on professional judgement not reducible to a computable rule. Absence means false.
sharedDecision	O	0..1	Boolean	True where the decision is intended to be made with the subject of care or their representative.
uncertainty	O	0..1	Text	Statement of material uncertainty in the decision or its evidence.
exceptionCondition	O	0..*	Reference	Exceptions applying to the decision.
logicProvenance	C	0..*	Reference	Evidence sources supporting the decision criteria. Mandatory where conformanceLevel is 4 and judgementBased is not true.

8.12 PathwayBranch

A PathwayBranch is an alternative route selected from a DecisionPoint. Its destination is recorded by exactly one branchTo relationship and any point at which it rejoins the main route by at most one rejoins relationship (9.2).

Table 18: PathwayBranch attributes

Attribute	Obl.	Mult.	Type	Description
condition	C	0..1	Expression	Condition under which the branch is selected. Mandatory where the branch belongs to an exclusive or inclusive decision point and isDefault is not true.
isDefault	O	0..1	Boolean	True where the branch is selected when no other condition applies. At most one branch of a decision point SHALL be the default.

8.13 Event

An Event is a clinically or operationally relevant occurrence that can influence pathway behaviour. An occurrence outside the pathway that serves as an anchor for a temporal constraint **SHALL** be represented as an Event with external set to true. An external event has no structural container (9.3) and participates in behavioural analysis only through its outgoing behavioural relationships (9.6). [\[CONF-HPDS-037\]](#)

Table 19: Event attributes

Attribute	Obl.	Mult.	Type	Description
eventType	M	1..1	Enumeration	clinical, operational, temporal, administrative, patient-initiated or other.
effect	C	0..*	Enumeration	initiates, modifies, interrupts, escalates, completes or terminates. Mandatory unless external is true and the event has no outgoing behavioural relationship.
trigger	O	0..1	Expression	Condition or occurrence constituting the event.
external	O	0..1	Boolean	True where the event occurs outside the pathway, such as an anchor for timing. Absence means false.

8.14 Role

A Role is a responsibility for activity, decision-making or coordination, defined independently of named individuals. Roles include patient and carer roles. Responsibilities are recorded by performs, decides and accountableFor relationships (9.2).

Table 20: Role attributes

Attribute	Obl.	Mult.	Type	Description
roleCategory	M	1..1	Enumeration	clinical, professional, administrative, coordination, patient, carer or other.
roleCode	O	0..*	TerminologyBinding	Bindings to role or occupation codes.
requiredCompetence	O	0..*	Text	Competence, registration or qualification required.

8.15 Service

A Service is a healthcare or care service that delivers or supports pathway elements. A Service in a pathway definition **SHALL NOT** identify a named provider organisation. Care settings are related to services by occursIn relationships. [\[CONF-HPDS-038\]](#)

Table 21: Service attributes

Attribute	Obl.	Mult.	Type	Description
serviceType	M	1..1	CodeableText	Type of service, including social care and voluntary or community sector services where they contribute to the pathway.
specialty	O	0..*	TerminologyBinding	Clinical specialty or service classification.

8.16 CareSetting

Table 22: CareSetting attributes

Attribute	Obl.	Mult.	Type	Description
settingType	M	1..1	CodeableText	home, care home, primary care, community care, outpatient, inpatient, diagnostic facility, emergency department, virtual or other.

8.17 InformationRequirement

An InformationRequirement specifies information needed to support an activity or decision. The elements that need it are identified by requires relationships. Every information requirement **SHALL** state its purpose. [\[CONF-HPDS-039\]](#)

Table 23: InformationRequirement attributes

Attribute	Obl.	Mult.	Type	Description
concept	M	1..1	Reference	ClinicalConcept or identified data element required.
purpose	M	1..1	Text	Why the information is required and what depends on it.
obligation	M	1..1	Enumeration	mandatory, optional or conditional.
obligationCondition	C	0..1	Expression	Condition under which the requirement applies. Mandatory where obligation is conditional.
expectedSource	O	0..*	CodeableText	Expected source, such as a role, service, the subject of care or a prior information product.
timeliness	O	0..1	TemporalConstraint	Required currency or availability of the information.
qualityRequirement	O	0..*	Text	Required completeness, accuracy, validity or other quality characteristic.
informationStandard	O	0..*	Reference	Applicable information standard or dataset specification.

8.18 InformationProduct

An InformationProduct specifies information generated through an activity, decision or event. The generating elements are identified by produces relationships.

Table 24: InformationProduct attributes

Attribute	Obl.	Mult.	Type	Description
productType	M	1..1	Enumeration	observation, assessment, report, referral, order, result, plan, discharge-information, decision-record or other.
concept	O	0..*	Reference	ClinicalConcepts conveyed.
informationStandard	O	0..*	Reference	Applicable information standard.
recipient	O	0..*	Reference	Roles or services that receive or are informed by the product.

NOTE A shared care plan maintained across services is represented as an InformationProduct with productType plan, produced and updated by the activities that contribute to it and required by those that rely on it.

8.19 ClinicalConcept

A ClinicalConcept is a unit of clinical or care meaning used within a pathway description. This document does not define clinical terminology.

Table 25: ClinicalConcept attributes

Attribute	Obl.	Mult.	Type	Description
conceptCategory	O	0..1	Enumeration	finding, disorder, procedure, observable, substance, medicinal-product, device, situation, social-context or other.
valueSet	O	0..1	Reference	Value set constraining permitted values, with version.

8.20 Outcome

An Outcome is an intended result of the pathway or of a component of it. A pathway definition **SHALL** contain at least one Outcome. An Outcome **SHALL NOT** be defined solely as completion of an activity or stage. Contributing elements are identified by supports relationships. Observed outcomes belong to pathway instances and are outside the scope of this document. [\[CONF-HPDS-040\]](#)

Table 26: Outcome attributes

Attribute	Obl.	Mult.	Type	Description
outcomeScope	M	1..1	Enumeration	pathway, stage or activity.
statement	M	1..1	CodeableText	Statement of the intended outcome.
timeframe	O	0..1	TemporalConstraint	Timeframe within which the outcome is expected.

8.21 Measure

A Measure is a defined measurable property associated with pathway performance or outcome achievement. The elements assessed are identified by assesses relationships.

Table 27: Measure attributes

Attribute	Obl.	Mult.	Type	Description
measureDomain	M	1..1	Enumeration	clinical-outcome, patient-experience, safety, timeliness, effectiveness, equity, utilisation, adherence or other.
definition	M	1..1	Expression	Definition of the measure, including numerator and denominator where relevant.
target	O	0..1	Text	Target or threshold.
measureReference	O	0..1	Reference	External measure specification.

8.22 Constraint

Table 28: Constraint attributes

Attribute	Obl.	Mult.	Type	Description
constraintType	M	1..1	Enumeration	temporal, resource, regulatory, information-governance, clinical, dependency or other.
appliesTo	M	1..*	Reference	Elements to which the constraint applies.
statement	M	1..1	Expression	Statement of the constraint.

Attribute	Obl.	Mult.	Type	Description
severity	O	0..1	Enumeration	mandatory, target or advisory.
temporal	C	0..1	TemporalConstraint	Temporal detail (Chapter 10). Mandatory where constraintType is temporal.

NOTE Requirements for consent, capacity, confidentiality and proportionate information sharing are recorded as constraints of type information-governance, expressed in business terms without reference to particular access-control mechanisms.

8.23 Exception

Exceptions **SHALL** be represented as Exception elements and **SHALL NOT** be recorded solely within narrative attributes of other elements. [\[CONF-HPDS-041\]](#)

Table 29: Exception attributes

Attribute	Obl.	Mult.	Type	Description
condition	M	1..1	Expression	Circumstance in which normal behaviour does not apply.
appliesTo	M	1..*	Reference	Elements affected.
alternativeBehaviour	M	1..1	Text	Behaviour applying instead.
alternativeRoute	O	0..1	Reference	Branch, behavioural node, exit point or pathway followed instead.
authorisingRole	O	0..*	Reference	Roles that MAY invoke the exception.

8.24 EvidenceSource

Table 30: EvidenceSource attributes

Attribute	Obl.	Mult.	Type	Description
sourceType	M	1..1	Enumeration	guidance, legislation, policy, research-evidence, consensus-statement, local-evidence or other.
citation	M	1..1	Text	Bibliographic citation.
locator	O	0..1	Text	URI or other locator.
sourceVersion	O	0..1	Text	Version or date of the source.
recommendationReference	O	0..1	Text	Specific recommendation or section.
strength	O	0..1	Text	Strength or grade as stated by the source.

8.25 InteroperabilityRequirement

InteroperabilityRequirement elements are specified in Table 31. Requirements for their content and derivation are given in Chapter 12.

Table 31: InteroperabilityRequirement attributes

Attribute	Obl.	Mult.	Type	Description
information	M	1..*	Reference	InformationProduct or InformationRequirement exchanged.
sender	M	1..1	Reference	Sending role or service.
receiver	M	1..*	Reference	Receiving roles or services.
trigger	M	1..1	Reference	Event or element whose occurrence or completion gives rise to the exchange.
interactionPattern	O	0..1	Enumeration	notification, request-response, query, publication, shared-record or other.
timing	O	0..1	TemporalConstraint	Required timing.
semanticRequirement	M	1..1	Text	Meaning that SHALL be preserved in the exchange.
informationStandard	O	0..*	Reference	Applicable national or international information standards.
accessRequirement	O	0..*	Reference	Constraints of type information-governance applying to the exchange (8.22).

8.26 Structured data types

8.26.1 TerminologyBinding

Table 32: TerminologyBinding structure

Component	Obl.	Mult.	Type	Description
bindingKind	M	1..1	Enumeration	concept, value-set or data-element.
system	M	1..1	Identifier	Code system, value set or information standard.
systemVersion	M	1..1	Text	Version of the semantic resource.
code	M	1..1	Text	Code, value set identifier or data element identifier.
display	O	0..1	Text	Display text.
bindingStrength	C	0..1	Enumeration	required, extensible, preferred or example. Mandatory where bindingKind is value-set.

8.26.2 Expression

Table 33: Expression structure

Component	Obl.	Mult.	Type	Description
narrative	M	1..1	Text	Human-readable statement. Always required, including where a computable expression is present.
expressionProfile	C	0..1	Reference	Expression profile (13.4). Mandatory where expression is present.
language	C	0..1	Text	Expression language. Mandatory where expression is present.
languageVersion	C	0..1	Text	Version of the expression language. Mandatory where expression is present.
expression	O	0..1	Text	Computable expression body.
concept	C	0..*	Reference	ClinicalConcepts on which the expression depends. Mandatory, with at least one entry, at Level 3 and above for the expressions of population inclusion and exclusion criteria and of entry and exit criteria (11.3).

8.26.3 TemporalConstraint and RecurrenceSpec

TemporalConstraint is specified in Chapter 10. RecurrenceSpec is specified in Table 34.

Table 34: RecurrenceSpec structure

Component	Obl.	Mult.	Type	Description
frequency	M	1..1	Duration	Interval between occurrences.
count	O	0..1	Integer	Number of occurrences.
until	O	0..1	Expression	Condition ending the recurrence.

8.26.4 IterationSpec

Table 35: IterationSpec structure

Component	Obl.	Mult.	Type	Description
terminationCondition	M	1..1	Expression	Condition that ends the repetition.
maximumIterations	O	0..1	Integer	Maximum number of repetitions.
minimumInterval	O	0..1	Duration	Minimum interval between repetitions.

8.26.5 ProvenanceRecord

Table 36: ProvenanceRecord structure

Component	Obl.	Mult.	Type	Description
origin	M	1..1	Enumeration	authored, derived-from-parent, derived-from-evidence, derived-from-representation, imported-from-catalogue or other.
author	M	1..*	Text	Authors.
date	M	1..1	Date	Date of authorship, derivation or import.
sourceReference	O	0..*	Reference	Evidence source, parent element, formal representation or source catalogue entry.

Component	Obl.	Mult.	Type	Description
approval	O	0..*	ApprovalRecord	Approvals applying specifically to the element.
note	O	0..1	Text	Explanatory note, including the nature of evidential support where the content is a synthesis.

8.26.6 ApprovalRecord

Table 37: ApprovalRecord structure

Component	Obl.	Mult.	Type	Description
approvalType	M	1..1	Enumeration	review, clinical, organisational or implementation.
approver	M	1..1	Text	Person approving.
approverRole	M	1..1	Text	Role in which approval is given.
approvingBody	O	0..1	Text	Body on whose behalf approval is given.
date	M	1..1	Date	Date of approval.
scope	O	0..1	Text	Parts of the description to which the approval applies.

8.26.7 ChangeRecord

Table 38: ChangeRecord structure

Component	Obl.	Mult.	Type	Description
version	M	1..1	Text	Version introduced by the change.
date	M	1..1	Date	Date of the change.
author	M	1..1	Text	Author of the change.
changeCategory	M	1..1	Enumeration	major, minor or patch (15.3).
summary	M	1..1	Text	Summary of the change.
rationale	M	1..1	Text	Reason for the change.
affectedElement	O	0..*	Identifier	Identifiers of affected elements.
boundResourceChange	C	0..*	Text	Change of version of a bound code system, value set or information standard. Mandatory where such a change is made (11.4).

8.26.8 Deviation

Table 39: Deviation structure

Component	Obl.	Mult.	Type	Description
identifier	M	1..1	Identifier	Persistent identifier of the deviation.
deviationType	M	1..1	Enumeration	added, omitted, modified, substituted or reordered.
parentElement	C	0..1	Reference	Element in the parent definition, with version. Mandatory unless deviationType is added.
localElement	C	0..1	Reference	Element in the variant or implementation. Mandatory unless deviationType is omitted.
changedAttribute	O	0..*	Text	Attributes modified.
rationale	M	1..1	Text	Reason for the deviation.
approval	O	0..*	ApprovalRecord	Approval of the deviation.
evidence	O	0..*	Reference	Supporting evidence sources.

8.26.9 Extension

Table 40: Extension structure

Component	Obl.	Mult.	Type	Description
url	M	1..1	Identifier	Persistent identifier resolving to the extension definition.
value	M	1..1	Any permitted by the extension definition	Value of the extension.
modifying	M	1..1	Boolean	Whether the extension changes the interpretation of the element that carries it.

8.26.10 GovernanceEvent

A GovernanceEvent records a change to the governance state of a published description (14.2).

Table 41: GovernanceEvent structure

Component	Obl.	Mult.	Type	Description
eventType	M	1..1	Enumeration	publication, lifecycle-change, authority-change, approval, review-outcome, supersession or other.
date	M	1..1	Date	Date of the event.
actor	M	1..1	Text	Person, role or organisation recording or authorising the event.
lifecycleStatus	C	0..1	Enumeration	Lifecycle state established by the event (14.5). Mandatory where eventType is publication, lifecycle-change or supersession.
authorityStatus	C	0..1	Enumeration	Authority status established by the event (16.1). Mandatory where eventType is publication or authority-change.
reviewDate	C	0..1	Date	Review date established by the event. Mandatory where eventType is publication or review-outcome.
approval	C	0..*	ApprovalRecord	Approvals given. Mandatory where eventType is approval.
clinicalAuthority	O	0..*	Text	Clinical or professional authority approving the content by this event.
supersededBy	C	0..1	Reference	Version that supersedes this version. Mandatory where eventType is supersession.
note	O	0..1	Text	Explanation, such as the outcome of a review.

8.26.11 RepresentationDiscrepancy

Table 42: RepresentationDiscrepancy structure

Component	Obl.	Mult.	Type	Description
identifier	M	1..1	Identifier	Persistent identifier of the discrepancy.
discrepancyType	M	1..1	Enumeration	contradiction or unresolved (8.10.1).
affectedElement	M	1..*	Reference	Elements of the definition affected.
description	M	1..1	Text	Description of the difference.
status	M	1..1	Enumeration	open or resolved.
resolution	C	0..1	Text	How the discrepancy was resolved. Mandatory where status is resolved.
date	M	1..1	Date	Date on which the discrepancy was recorded or its status last changed.

8.27 PathwayImplementation

A PathwayImplementation describes how an organisation or health system realises a pathway definition. It is not part of the technology-neutral pathway definition. An implementation description **SHALL** contain exactly one PathwayImplementation element. The authority status of an implementation is its own. Clinical approval of the realised definition does not confer clinical approval on the implementation; the requirements of Table 54 apply to the implementation's own approvals and clinical authority. [\[CONF-HPDS-042\]](#)

Table 43: PathwayImplementation attributes

Attribute	Obl.	Mult.	Type	Description
realises	M	1..1	Reference	Pathway definition and version realised.
version	M	1..1	Text	Version of the implementation description.
lifecycleStatus	M	1..1	Enumeration	Current lifecycle state of the implementation description (14.5). Governance-state attribute (14.2).
authorityStatus	M	1..1	Enumeration	Current authority status of the implementation description (16.1). Governance-state attribute (14.2).
intendedUse	M	1..*	Enumeration	Purposes for which the implementation description is intended to be relied upon (16.2).
owner	M	1..1	Text	Accountable owner of the implementation description.
implementingOrganisation	M	1..*	Text	Organisations responsible for the implementation.

Attribute	Obl.	Mult.	Type	Description
clinicalAuthority	O	0..*	Text	Clinical or professional authority approving the clinical content of the implementation, as recorded when its content was fixed. Evaluated with governance events against Table 54 for the implementation's current authority status (16.1, V06).
approval	O	0..*	ApprovalRecord	Approvals of the implementation recorded when its content was fixed. Evaluated with governance events against Table 54 (16.1, V06).
mapping	M	1..*	ImplementationMapping	Mappings from definition elements to local implementation elements (8.28).
deviation	C	0..*	Deviation	Differences from the realised pathway definition. Mandatory for every difference (13.2).
automationArtefact	O	0..*	ImplementationArtefact	Technical artefacts that realise aspects of the pathway (8.28, 16.3).
effectiveDate	C	0..1	Date	Date from which the implementation applies. Mandatory where lifecycleStatus is published or later.
reviewDate	C	0..1	Date	Current date by which the implementation is due for review. Mandatory under the same condition as effectiveDate. Governance-state attribute (14.2).
modifyingExtension	C	0..*	Identifier	Identifiers of modifying extensions used. Mandatory where any modifying extension is used.
changeHistory	C	0..*	ChangeRecord	History of published changes. Mandatory for every version after the first published version.
governanceEvent	C	0..*	GovernanceEvent	Governance events recorded for this version, in non-decreasing date order. At least one event of type publication is mandatory where lifecycleStatus is published or any later state (14.2). Governance-state attribute.

8.28 Implementation data types

8.28.1 ImplementationMapping

Table 44: ImplementationMapping structure

Component	Obl.	Mult.	Type	Description
definitionElement	M	1..1	Reference	Element in the realised definition version.
implementationKind	M	1..1	Enumeration	provider-service, location, workforce-group, application, application-function, interface, order-set, worklist, local-procedure or other.
implementationIdentifier	M	1..1	Text	Local identifier of the implementation element.
label	O	0..1	Text	Display label.

8.28.2 ImplementationArtefact

Table 45: ImplementationArtefact structure

Component	Obl.	Mult.	Type	Description
identifier	M	1..1	Identifier	Identifier of the artefact.
artefactType	M	1..1	Enumeration	workflow-definition, case-model, decision-model, decision-support-rule, fhir-artefact, system-configuration, order-set or other.
locator	M	1..1	Text	URI or other locator of the artefact.
artefactVersion	M	1..1	Text	Version of the artefact.
standard	O	0..*	Reference	Standard in which the artefact is expressed, such as BPMN, DMN or HL7 FHIR.
realisesElement	O	0..*	Reference	Definition elements realised by the artefact.
localisesRepresentation	O	0..1	Reference	FormalRepresentation of the realised definition from which the artefact was localised (8.10).
drivesAutomatedActivity	M	1..1	Boolean	Whether the artefact configures or drives automated clinical or operational activity.
assuranceEvidence	C	0..*	Text	Reference to clinical safety or other assurance evidence. Mandatory where drivesAutomatedActivity is true (16.3).

8.29 Reference points for pathway instances

Systems that record pathway instances **SHOULD** record, for each relevant instance event, the identifier and version of the pathway definition, the identifier and version of the pathway implementation where applicable, and the identifier of the element to which the event corresponds. Where a system assesses the relationship between an instance and a pathway, it **SHOULD** record the result as consistent, deviating or indeterminate. Whether a deviation was clinically warranted is a clinical judgement outside the scope of this document.

NOTE Identifiers persist through changes of name (Chapter 15), so instance data recorded against an earlier version remain interpretable. The variance record described in some definitions of clinical pathways [5] belongs at instance level and is supported by these reference points rather than by the pathway definition.

9 Relationships and graph semantics

9.1 Relationship representation

Relationships **SHALL** be represented explicitly as identified Relationship elements carrying the common attributes in 8.3 and the attributes in Table 46. Behaviour **SHALL NOT** be inferred from presentational order or from structural relationships. A relationship type not listed in Table 47 **SHALL** be defined by an extension (13.6) and identified by an absolute URI. [\[CONF-HPDS-043\]](#)

Table 46: Relationship attributes

Attribute	Obl.	Mult.	Type	Description
relationshipType	M	1..1	Enumeration	Type from Table 47, or a type defined by an extension.
source	M	1..1	Reference	Source element.
target	M	1..1	Reference	Target element. A target in another pathway description SHALL be version qualified.
condition	O	0..1	Expression	Condition under which the relationship applies.
joinRule	C	0..1	Enumeration	all, any or count. Mandatory on each precedes relationship whose target has more than one incoming precedes relationship; all such relationships to one target SHALL carry the same value.
joinCount	C	0..1	Integer	Number of incoming routes required. Mandatory where joinRule is count.
iteration	C	0..1	IterationSpec	Mandatory where relationshipType is iterates.

9.2 Relationship types

Permitted relationship types, sources, targets and cardinalities are given in Table 47. The cardinality column states how many targets a source can have and, after the semicolon, how many sources a target can have.

Table 47: Relationship types

Type	Category	Permitted source	Permitted target	Cardinality	Semantics
contains	Structural	HealthcarePathway, PathwayStage	PathwayStage, Activity, DecisionPoint, Event	0..*; 1..1	Structural containment. Each contained element has exactly one container.
includes	Structural	Activity; HealthcarePathway or PathwayStage at Level 1 only	HealthcarePathway (sub-pathway, version qualified)	0..1 from an Activity, otherwise 0..*; 0..*	Inclusion of a sub-pathway by reference. From an Activity, the activity is realised by the sub-pathway (9.3). The sub-pathway is not copied.
startsAt	Behavioural	EntryPoint	Behavioural node	1..1; 0..*	Initial behavioural node for an entry point.
precedes	Behavioural	Behavioural node	Behavioural node, ExitPoint	0..*; 0..*	Target commences after source completes, subject to any relationship condition and to the join rule of the target (9.8).
selects	Behavioural	DecisionPoint	PathwayBranch	2..*; 1..1	Decision selects among branches. Each branch belongs to one decision point.
branchTo	Behavioural	PathwayBranch	Behavioural node, ExitPoint	1..1; 0..*	Destination of a branch.
rejoins	Behavioural	PathwayBranch	Behavioural node	0..1; 0..*	Node at which a branch rejoins the main route.
triggers	Behavioural	Event	Behavioural node, InteroperabilityRequirement	0..*; 0..*	Occurrence commences or activates the target.
interrupts	Behavioural	Event	PathwayStage, Activity	0..*; 0..*	Occurrence suspends or ends the target.
escalatesTo	Behavioural	Event, DecisionPoint; HealthcarePathway at Level 1 only	Behavioural node, ExitPoint, HealthcarePathway (version qualified)	0..*; 0..*	Escalation route.
iterates	Behavioural	Activity, PathwayStage	Activity, PathwayStage	0..1; 0..*	Repetition of the route from the target to the source, subject to the iteration specification.
waitsFor	Behavioural	Activity, PathwayStage	Event	0..*; 0..*	Source does not complete until the event occurs.
transitionsTo	Behavioural	ExitPoint; HealthcarePathway at Level 1 only	HealthcarePathway (version qualified)	0..1; 0..*	Transfer into another pathway. From a HealthcarePathway the cardinality is 0..*.

Type	Category	Permitted source	Permitted target	Cardinality	Semantics
dependsOn	Dependency	Activity, DecisionPoint	Activity, InformationProduct	0..*; 0..*	Dependency on the result of the target, independent of order.
requires	Dependency	Activity, DecisionPoint	InformationRequirement	0..*; 1..*	Information needed.
produces	Dependency	Activity, DecisionPoint, Event	InformationProduct	0..*; 1..*	Information generated.
performs	Responsibility	Role	Activity	0..*; 0..*	Role responsible for performance.
decides	Responsibility	Role	DecisionPoint	0..*; 1..*	Role responsible for a decision.
accountableFor	Responsibility	Role	HealthcarePathway, PathwayStage, Activity	0..*; 0..*	Role accountable for coordinating or overseeing the target, independently of who performs it.
provides	Responsibility	Service	PathwayStage, Activity	0..*; 0..*	Service delivers or supports the target.
occursIn	Context	PathwayStage, Activity, Service	CareSetting	0..*; 0..*	Setting in which the source occurs or operates.
supports	Outcome	PathwayStage, Activity	Outcome	0..*; 0..*	Source contributes to the outcome.
assesses	Outcome	Measure	Outcome, HealthcarePathway, PathwayStage, Activity	1..*; 0..*	Measure assesses the target.
specialises	Derivation	HealthcarePathway	HealthcarePathway (version qualified)	0..1; 0..*	Source is a variant of the target.
realises	Derivation	PathwayImplementation	HealthcarePathway (version qualified)	1..1; 0..*	Implementation realises the definition.

NOTE 1 In this table “behavioural node” means PathwayStage, Activity, DecisionPoint or Event (3.23).

NOTE 2 The minimum of 1 in “0..*; 1..*” for requires, produces and decides applies at Level 2 and above; at Level 1 these relationships may be incomplete.

NOTE 3 At Level 1 a transfer, escalation or inclusion of another pathway can be recorded from the HealthcarePathway as a whole, because the exit point, event or activity from which it arises might not yet be described.

At Level 2 and above, the source of every transitionsTo and escalatesTo relationship **SHALL** be an ExitPoint, Event or DecisionPoint (V55), and the source of every includes relationship **SHALL** be an Activity (V60).

[CONF-HPDS-044]

9.3 Structural containment and inclusion

At Level 2 and above, every PathwayStage, Activity and DecisionPoint, and every Event whose external attribute is not true, **SHALL** have exactly one structural container within a pathway definition. An Event whose external attribute is true **SHALL NOT** have a structural container. Structural relationships establish behavioural reachability only as stated in 9.6. [CONF-HPDS-045]

A sub-pathway **SHALL** be included by an includes relationship and **SHALL NOT** be copied into the including definition. The relationship **SHALL** identify the version of the sub-pathway or a permitted version range. The content of an included sub-pathway is validated within its own pathway definition. [CONF-HPDS-046]

For behavioural analysis of the including definition, an Activity that includes a sub-pathway is a single behavioural node. It commences as any other activity and is treated as complete when the included sub-pathway reaches one of its exit points. The elements of the included definition are not part of the including behavioural graph and are not evaluated for reachability, containment or cycles within it. Analyses that span the inclusion boundary **MAY** use the entry and exit points of the included version. An includes relationship from a HealthcarePathway or PathwayStage, permitted at Level 1 only, records that the sub-pathway is used without placing it in the behavioural graph.

9.4 Behavioural patterns

The behavioural patterns in Table 48 **SHALL** be represented as specified. [CONF-HPDS-047]

Table 48: Representation of behavioural patterns

Pattern	Representation
Sequential flow	precedes relationship from the earlier to the later element.
Parallel split	Two or more precedes relationships from one source. Every route is followed, subject to its relationship condition, if any. Alternative routes are represented by a DecisionPoint.

Pattern	Representation
Parallel join	Two or more precedes relationships to one target, each with joinRule all, any or count, and joinCount where the rule is count.
Conditional branch	DecisionPoint selects two or more PathwayBranch elements; each branch has one branchTo relationship and a condition, or is the single default branch.
Rejoining branch	PathwayBranch with a rejoins relationship.
Unordered (adaptive) work	PathwayStage with ordering unordered containing the available elements, with prerequisites, entry conditions or triggers on those elements and a completion condition on the stage.
Entry into an ordered stage	The initial nodes of the stage commence when the stage commences (9.6). No relationship from a stage to its own contents is used.
Optional or discretionary activity	Activity.optional true and, within an ordered stage, a condition on the governing relationship where needed.
Iteration	iterates relationship with an IterationSpec.
Waiting	waitsFor relationship, or a temporal constraint of type minimum-interval or deadline.
Escalation	escalatesTo relationship from an Event or DecisionPoint.
Interruption	Event with effect interrupts and an interrupts relationship to the affected element.
Termination	ExitPoint reached by precedes or branchTo, or Event with effect terminates and a precedes relationship to an ExitPoint.
Transition to another pathway	ExitPoint with exitType transfer and a transitionsTo relationship.
Use of a sub-pathway	includes relationship from the Activity realised by the sub-pathway; at Level 1 only, from the including pathway or stage.
Care coordination	accountableFor relationship from the coordinating Role to the pathway, stage or activity coordinated.

9.5 Iteration

Repetition **SHALL** be represented only by iterates relationships, each carrying an IterationSpec with a termination condition. The source of an iterates relationship is the last element of the repeated route and its target is the first. [CONF-HPDS-048]

The cycle-detection graph of a pathway definition has as nodes its behavioural nodes, pathway branches and exit points, and as edges:

- every precedes, selects, branchTo, rejoins, triggers and escalatesTo relationship whose source and target are both elements of the definition;
- an implicit edge from each ordered stage to each of its initial nodes, and from each unordered stage to each node it directly contains.

Iterates, interrupts, waitsFor and non-behavioural relationships are not edges of the cycle-detection graph. The cycle-detection graph **SHALL** be acyclic (V30). The source of every iterates relationship **SHALL** be reachable from its target in the cycle-detection graph (V68). [CONF-HPDS-049]

NOTE Including selects and the implicit stage edges ensures that a cycle through a decision point and its branches, or through a stage and its contents, is detected.

9.6 Behavioural reachability

Reachability is defined over the elements of one pathway definition. The set of reachable elements is the smallest set that satisfies the following rules:

- the target of every startsAt relationship is reachable;
- where the source of a precedes, selects, branchTo, rejoins, triggers or escalatesTo relationship is reachable, or is an Event whose external attribute is true, its target is reachable if the target is an element of the definition;
- where an ordered stage is reachable, its initial nodes (3.60) are reachable;
- where an unordered stage is reachable, every node it directly contains is reachable.

In applying rule c), an incoming relationship counts against a node's being initial only if its source is directly contained in the same stage, or is a PathwayBranch selected by a DecisionPoint directly contained in the same stage. Nested stages are reachable on the same terms as other nodes and then make their contents reachable by rules c) and d). Optional activities are reachable on the same terms as other activities; being

optional affects completion only (8.8). An Activity that includes a sub-pathway is treated as a single node (9.3). Structural containment, other than as stated in rules c) and d), and iterates, interrupts, waitsFor, dependency, responsibility, context, outcome and derivation relationships do not make any element reachable.

At Level 2 and above:

- e) every behavioural node other than an Event whose external attribute is true **SHALL** be reachable (V28); [CONF-HPDS-050]
- f) for each EntryPoint, applying the rules above with only the startsAt relationship of that entry point in rule a), at least one ExitPoint **SHALL** be reachable (V29); [CONF-HPDS-051]
- g) the target of every waitsFor relationship **SHALL** be an Event whose external attribute is true or a reachable Event (V67). [CONF-HPDS-052]

An Event whose external attribute is true and which has no outgoing behavioural relationship, such as an event used only as a temporal anchor, takes no part in reachability. Reachability is a structural property of the description; it does not indicate that an element is performed for every subject of care.

9.7 Process execution

The behavioural graph expresses intended business behaviour with the descriptive semantics stated in 9.8. It is not an executable workflow definition. A BPMN, CMMN, DMN, FHIR or other computable model **MAY** be referenced from a pathway definition as a formal representation (8.10), or linked to a pathway implementation as an ImplementationArtefact (8.28.2). Execution semantics (1.2) are outside the scope of this document, and no conformance level of this document implies that a pathway can be executed.

9.8 Descriptive behavioural semantics

The behavioural relationships of this document have the descriptive meanings in Table 49. A descriptive meaning states what is intended to happen relative to other elements; it does not state how, when or by what mechanism a system brings it about. Each construct is retained because it supports at least one of the analyses listed.

Table 49: Descriptive behavioural semantics

Construct	Descriptive meaning	Business purpose	Analyses supported
startsAt	The target is where the pathway begins for subjects entering by that entry point.	Distinguish routes of entry.	Reachability; comparison of entry routes.
precedes	The target commences after the source completes; with several outgoing precedes, every route is followed, each subject to its condition.	Intended order and concurrency.	Reachability; termination; dependency and impact analysis; mapping to sequence flow.
joinRule (all, any, count)	The target commences after all, any one, or the stated number of its incoming routes have completed.	Synchronisation after concurrent work.	Comparison of variants; mapping to gateways.
Relationship condition	The relationship applies only when the condition holds.	Routes that apply only to some subjects of care without a distinct decision.	Comparison; Level 4 computability.
selects, branchTo, rejoins	A decision point selects one (exclusive), one or more (inclusive) or an event-determined (deferred) branch; each branch leads to its destination and can rejoin the main route.	Explicit, accountable choice between alternatives.	Reachability; decision responsibility; variation analysis.
iterates	The route from target to source is repeated until the termination condition holds.	Repeated reviews, cycles of treatment.	Cycle validation; comparison.
waitsFor	The source does not complete until the event occurs.	Waiting for results, decisions or external occurrences.	Dependency analysis; timing analysis.
triggers	The occurrence of the event commences or activates the target.	Event-driven and adaptive care.	Reachability; mapping to event listeners.
interrupts	The occurrence of the event suspends or ends the target.	Interruption by deterioration or patient choice.	Safety and exception analysis.
escalatesTo	The occurrence or decision leads to the escalation target.	Escalation routes.	Reachability; comparison; safety analysis.
transitionsTo	Leaving by the exit point leads into the target pathway.	Connection between pathways.	Catalogue navigation; cross-pathway analysis.
includes (from an Activity)	The activity is realised by the sub-pathway.	Reuse without copying.	Modular analysis; impact of changes to sub-pathways.

Construct	Descriptive meaning	Business purpose	Analyses supported
Ordered stage	Initial nodes commence when the stage commences; the stage completes when its completion conditions hold or its terminal nodes have completed.	Grouping with intended internal order.	Reachability; presentation; mapping to sub-processes.
Unordered stage	Every contained node is available while the stage is active; the stage completes when its completion condition holds.	Adaptive, need-driven work.	Reachability; presentation; mapping to case stages.
Activity.optional	The activity need not be performed for the stage to complete.	Discretionary work.	Completion analysis; comparison.

These semantics support the following analyses: structural validation (V17 to V33, V55, V60, V64, V65, V67, V68); comparison of pathways and of variants with their parents (13.2); dependency and impact analysis; derivation of interoperability requirements (Chapter 12); and mapping to formal representations (Appendices C and D). Descriptive semantics do not define run-time instance state, the time at which an element commences, the assignment of work, or the treatment of failures, which are execution semantics (1.2).

9.9 Graph validation outcomes

For a declared conformance level, a behavioural graph is invalid if any validation rule of severity Error that applies at that level fails. At Level 1 the graph **MAY** be incomplete: minimum relationship cardinalities, containment, reachability, termination and cycle rules are not evaluated, but relationships that are present **SHALL** be of permitted types (V17). Rules of severity Warning identify conditions that require human review; they do not prevent conformance. Table 50 summarises the outcomes. [\[CONF-HPDS-053\]](#)

Table 50: Graph validation outcomes by level

Condition	Level 1	Levels 2 to 4
Relationship with a source or target type not permitted, or exceeding a maximum cardinality	Invalid (V17)	Invalid (V17)
Minimum relationship cardinalities not met	Permitted (incomplete)	Invalid (V17, V20 to V27)
Element without the required container; external event with a container	Not evaluated	Invalid (V18)
Behavioural node not reachable; entry point from which no exit point is reachable	Not evaluated	Invalid (V28, V29)
Cycle in the cycle-detection graph; iterates relationship without a repeated route	Not evaluated	Invalid (V30, V68)
waitsFor target neither external nor reachable	Not evaluated	Invalid (V67)
transitionsTo or escalatesTo from the pathway; includes from a pathway or stage	Permitted	Invalid (V55, V60)
Ordered stage with more than one initial node	Not evaluated	Warning for review (V64)
Reachable node directly contained in the pathway with no route to an exit point	Not evaluated	Warning for review (V65)

10 Temporal representation

10.1 TemporalConstraint

Temporal constraints **SHALL** be represented using the TemporalConstraint structure in Table 51. Every temporal constraint **SHALL** have a narrative statement. At Level 4, temporal constraints **SHALL** also carry a computable expression conforming to the declared expression profile. [\[CONF-HPDS-054\]](#)

Table 51: TemporalConstraint structure

Component	Obl.	Mult.	Type	Description
temporalType	M	1..1	Enumeration	maximum-wait, minimum-interval, target-interval, expected-duration, deadline, recurrence, relative-offset or sequence.
anchor	C	0..1	Reference	Anchor event or element. Mandatory for every temporal type except expected-duration and recurrence.
anchorPoint	C	0..1	Enumeration	start or end. Mandatory where the anchor is an element with duration.
minimum	O	0..1	Duration	Lower bound.
maximum	O	0..1	Duration	Upper bound.
target	O	0..1	Duration	Target value.
recurrence	C	0..1	RecurrenceSpec	Mandatory where temporalType is recurrence.
sequenceRelation	C	0..1	Enumeration	before, after, not-before or not-after. Mandatory where temporalType is sequence.
clinicalSignificance	O	0..1	Text	Clinical rationale or consequence of breach.
statement	M	1..1	Expression	Narrative statement and, where applicable, computable expression.

10.2 Relative time

Temporal constraints that depend on a prior occurrence **SHALL** be expressed relative to an anchor rather than as calendar dates. [\[CONF-HPDS-055\]](#)

NOTE Clinical timing requirements are commonly anchored to a referral, a result, a procedure, a discharge or a birth.

10.3 Consistency

A pathway definition **SHOULD NOT** contain temporal constraints that cannot be satisfied together. A producer that detects such constraints **SHALL** report a warning (Appendix A). [\[CONF-HPDS-056\]](#)

11 Semantic binding

11.1 General

This document does not define a clinical terminology. It enables binding to external semantic resources, such as SNOMED CT [10], ICD, national data dictionaries and information standards, without requiring any one of them.

11.2 Binding requirements

A terminology binding **SHALL** conform to 8.26.1 and **SHALL** identify the semantic resource, its version and the code, value set or data element identifier. Value set bindings **SHALL** state their binding strength. [CONF-HPDS-057]

11.3 Local terms

Local terms **SHALL** be identified by setting localTerm to true and **SHALL NOT** be represented as codes from an external semantic resource. At Level 3 and above, the clinical concepts on which population inclusion and exclusion criteria and entry and exit criteria depend **SHALL** be identified by the concept component of their Expressions (8.26.2), with at least one concept for each criterion. Every ClinicalConcept referenced by such an Expression, or by the concept attribute of a mandatory InformationRequirement, **SHALL** be bound to a versioned semantic resource or declared local. Concepts **SHALL NOT** be identified only by inference from narrative text. [CONF-HPDS-058]

11.4 Change in bound resources

A change to the version of a bound semantic resource **SHALL** be recorded in the change record of the version that introduces it. Producers **SHOULD** report bindings to versions that the publisher has withdrawn or superseded, where that information is available. [CONF-HPDS-059]

11.5 Binding report

A Level 3 or Level 4 conformance claim **SHALL** include a binding report. For the ClinicalConcepts in scope of V39 the report **SHALL** state: the number bound to external semantic resources, with the resources and versions used; the number declared local; and the identifiers of the local concepts. [CONF-HPDS-060]

Level 3 demonstrates that each in-scope concept is either bound to an identified, versioned external resource or explicitly declared local. It does not demonstrate that bindings are clinically correct, that independent parties will interpret local concepts identically, or that descriptions using different resources are semantically interoperable.

NOTE The reference validator in the release package generates a binding report (Appendix I).

12 Interoperability requirements

An InteroperabilityRequirement (8.25) states a business information exchange on which a pathway depends. It **SHALL** identify the information, sender, receiver, trigger and semantic requirement. It **SHALL NOT** specify endpoints, message structures, transport protocols, APIs or products as pathway content; technical realisation belongs to pathway implementations. [\[CONF-HPDS-061\]](#)

Where an information product generated by an element provided by one service is required by an element provided by a different service, the pathway definition **SHOULD** include an InteroperabilityRequirement covering that dependency. The absence of such a requirement is a warning (Appendix A).

13 Specialisation, variation and profiles

13.1 Specialisation

A pathway definition **MAY** specialise another pathway definition, and specialisation **MAY** be repeated: a national or professional reference pathway **MAY** be specialised regionally and organisationally before being realised in one or more implementations. A pathway variant **SHALL** reference its parent and parent version through the parent attribute and a specialises relationship. [CONF-HPDS-062]

13.2 Inheritance and deviation

An inherited element that retains the same meaning **SHALL** retain its parent identifier. A new element, or an inherited element whose meaning changes materially, **SHALL** receive a new identifier. Every added, omitted, modified, substituted or reordered element **SHALL** be recorded as a Deviation (8.26.8) with its rationale and, where applicable, approval and evidence. In a pathway variant, deviations are recorded in the deviation attribute of the HealthcarePathway (8.4); their parentElement references **SHALL** be qualified by the parent version. [CONF-HPDS-063]

At Level 2 and above, the deviations of a variant are complete when, comparing the elements of the variant with those of the referenced parent version by identifier (excluding the HealthcarePathway element itself):

- every identifier present in the parent but not in the variant is the parentElement of exactly one Deviation of type omitted or substituted;
- every identifier present in the variant but not in the parent is the localElement of exactly one Deviation of type added or substituted;
- every identifier present in both whose elements differ in any attribute other than provenance is the parentElement of exactly one Deviation of type modified or reordered, whose changedAttribute lists the differing attributes.

In this comparison, a reference to the HealthcarePathway element of the parent and a reference to the HealthcarePathway element of the variant are treated as equal.

Relationship elements are compared in the same way. Completeness is validated by V61. The same recording requirement applies to a PathwayImplementation in relation to the definition it realises, in its deviation attribute (8.27), including a change to the permitted outcomes of a decision point (8.11); because implementations add local elements of other kinds, completeness of implementation deviations is reviewed rather than computed (V47).

13.3 Domain profiles

A domain profile **MAY** tighten cardinalities and obligations, restrict value sets, require particular bindings and define extensions. It **SHALL NOT** relax any requirement of this document and **SHALL NOT** redefine the meaning of any core entity, attribute or relationship. A description that conforms to a domain profile **SHALL** also conform to this document. [CONF-HPDS-064]

NOTE Candidate domain profiles include diagnostic, surgical, maternity, medicines, chronic disease, care coordination, screening, and urgent and emergency care pathways.

13.4 Expression profiles

A Level 4 conformance claim **SHALL** identify at least one expression profile. An expression profile **SHALL** define the expression language and version, the evaluation context, data access conventions, terminology operations, the semantics of missing data and the permitted result types. This document does not mandate a single expression language. [CONF-HPDS-065]

NOTE Research on computer-interpretable guidelines found substantial common ground between formalisms but limited operational adoption of any one of them [13]. Expression profiles allow computable content without committing the core model to a single formalism.

13.5 Profile identification

A domain profile or expression profile **SHALL** have a persistent identifier, a version and a statement of the version of this document on which it is based. [CONF-HPDS-066]

13.6 Extensions

Extensions **SHALL** conform to 8.26.9. A non-modifying extension **SHALL NOT** change the interpretation of core content. Every modifying extension used in a description **SHALL** be listed in the modifyingExtension attribute of the HealthcarePathway or PathwayImplementation. [CONF-HPDS-067]

The content affected by a modifying extension is the element that carries it and, where that element is a PathwayStage, every element structurally contained in it, directly or indirectly; where that element is the HealthcarePathway or PathwayImplementation, the affected content is the whole description.

A consumer that does not support a modifying extension:

- SHALL NOT** interpret affected content for any purpose that depends on its meaning, at any conformance level, including semantic validation, comparison, analysis, derivation of interoperability requirements, presentation as guidance and evaluation of expressions; [CONF-HPDS-068]
- MAY** store, transmit and display affected content, provided that each presentation of it indicates that its meaning is modified by an unsupported extension;
- SHALL** preserve the extension and the affected content without alteration when storing or re-serialising the description; [CONF-HPDS-069]
- SHALL** report the unsupported modifying extensions and the identifiers of the affected elements whenever it reports validation or analysis results. [CONF-HPDS-070]

13.7 Expression coverage at Level 4

Table 52 lists every attribute whose type is Expression, or which contains an Expression, and states whether a computable expression is required at Level 4. At every level an Expression **SHALL** have a narrative. Where an attribute is marked Mandatory, every occurrence of it at Level 4 **SHALL** carry a computable expression conforming to a declared expression profile (V44). Where it is marked Optional, a computable expression **MAY** be given. The professional-judgement exception applies to the decision criteria of a DecisionPoint whose judgementBased attribute is true and to the conditions of every branch that decision point selects, and to no other attribute. Level 4 conformance does not imply that the pathway, or any part of it, is executable (5.2). [CONF-HPDS-071]

Table 52: Expression-bearing attributes at Level 4

Attribute	Computable expression at Level 4	Notes
Population.inclusionCriterion	Mandatory	
Population.exclusionCriterion	Mandatory	
Population.stratification	Optional	Describes variation within the population; selects no element.
EntryPoint.criterion	Mandatory	
ExitPoint.criterion	Mandatory	
PathwayStage.entryCondition	Mandatory	
PathwayStage.completionCondition	Optional	Completion is observed rather than selected (9.8).
Activity.prerequisite	Mandatory	
Activity.completionCriterion	Optional	As for stage completion.
DecisionPoint.decisionCriterion	Mandatory unless judgementBased is true	Logic provenance is required where judgementBased is not true (V45).
PathwayBranch.condition	Mandatory unless the decision point that selects the branch has judgementBased true	The judgement exception covers the decision criteria and all branch conditions of the same decision point.
Relationship.condition	Mandatory	
Event.trigger	Optional	Events are occurrences; their recognition is outside scope.
InformationRequirement.obligationCondition	Mandatory	
Measure.definition	Optional	Measures can be computed under separate specifications (measureReference).

Attribute	Computable expression at Level 4	Notes
Constraint.statement, where constraintType is temporal	Mandatory	Through the TemporalConstraint statement.
Constraint.statement, other constraint types	Optional	
Exception.condition	Mandatory unless the exception has at least one authorisingRole	An exception invoked by the judgement of an authorising role MAY remain narrative.
TemporalConstraint.statement, wherever it occurs	Mandatory	Includes InformationRequirement.timeliness, Outcome.timeframe and InteroperabilityRequirement.timing.
RecurrenceSpec.until	Mandatory	
IterationSpec.terminationCondition	Mandatory	

14 Governance, provenance and lifecycle

14.1 Governance metadata

Every pathway definition and pathway implementation **SHALL** record owner, version, lifecycle status, authority status and intended use. From the published state onward, effective and review dates **SHALL** also be recorded; the effective date is content and the review date is a governance-state attribute (14.2). Approvals **SHALL** be recorded as ApprovalRecords (8.26.6), and clinical authority as clinicalAuthority, in the content or in governance events, so that the requirements of 16.1 are met for the current authority status. A pathway definition published for use outside the owning organisation **SHOULD** record the rights under which it **MAY** be reused and specialised. [CONF-HPDS-072]

14.2 Content immutability and governance state

The attributes of a pathway description are divided into content and governance state. The governance-state attributes are lifecycleStatus, authorityStatus, reviewDate and governanceEvent of the HealthcarePathway or PathwayImplementation, and discrepancy of each FormalRepresentation. All other attributes of all elements are content.

Once a pathway description has been published, its content **SHALL NOT** be modified; any change to content **SHALL** produce a new version (Chapter 15). Its governance state **MAY** change after publication, but only by appending a GovernanceEvent (8.26.10) that records the change or, for representation discrepancies, by adding a RepresentationDiscrepancy or changing the status of an existing one to resolved. Existing governance events and discrepancy records **SHALL NOT** be removed, and governance events **SHALL NOT** be altered. [CONF-HPDS-073]

A GovernanceEvent of type publication **SHALL** be recorded when a description is published; it establishes the lifecycle status, authority status and review date at publication. Governance events **SHALL** be recorded in non-decreasing date order. Each of lifecycleStatus, authorityStatus and reviewDate **SHALL** equal the value established by the latest governance event that sets it (V62). [CONF-HPDS-074]

Historical versions **SHALL** remain interpretable and **SHALL** remain retrievable while they are referenced by implementations, catalogues or other pathway definitions. Catalogue manifests have no governance-state attributes; a published catalogue manifest version is wholly immutable. [CONF-HPDS-075]

14.3 Provenance

Provenance **SHALL** be recorded for the pathway description as a whole and for individual elements whose origin differs materially from that of the whole description. Content imported from a tabular catalogue **SHALL** carry a ProvenanceRecord with origin imported-from-catalogue. Content derived from a formal representation **SHALL** carry a ProvenanceRecord with origin derived-from-representation. [CONF-HPDS-076]

14.4 Change history

Every version after the first published version **SHALL** carry a ChangeRecord (8.26.7) that states the version, date, author, change category, summary and rationale, identifies affected elements where applicable, and records any change of bound resource version. [CONF-HPDS-077]

14.5 Lifecycle

Pathway descriptions **SHALL** use the lifecycle states and permitted transitions in Table 53. Organisations **MAY** use local lifecycle terminology provided each local state maps to exactly one state in Table 53. Transitions after publication are recorded as governance events of type lifecycle-change or supersession (14.2). [CONF-HPDS-078]

Table 53: Lifecycle states

State	Meaning	Permitted next states
draft	Under preparation.	review
review	Under formal review.	draft, approved
approved	Approved for publication.	published
published	Released for use.	under-review, superseded, retired

State	Meaning	Permitted next states
under-review	Subject to scheduled or triggered review. Content is unchanged; any resulting change produces a new version in the draft state.	published, superseded, retired
superseded	Replaced by a later version.	retired
retired	No longer in use; retained for historical interpretation.	none

NOTE Implementation is not a lifecycle state of a pathway definition. One definition may have zero, one or many implementations at the same time.

14.6 Historical and current state

The current governance state of a version is given by its governance-state attributes. The governance state of a version on a date D is determined by taking the governance events dated on or before D, in collection order, and applying each in turn to the state established by the publication event. The approvals and clinical authority in force on date D are those recorded in the content and in governance events dated on or before D.

A consumer **SHALL** use the current governance state when presenting a description. A consumer **SHOULD** use the historical governance state when interpreting instance data recorded against the version on a past date (8.29). Where a version has been superseded, a consumer **SHALL** present the supersession and the superseding version, if known. [\[CONF-HPDS-079\]](#)

15 Identification and versioning

15.1 Identifiers

Persistent identifiers **SHALL** be assigned to pathway descriptions, catalogue entries, formal representations and all elements, including stages, activities, decision points, outcomes, measures, relationships and reusable components. Identifiers **SHALL NOT** be derived from display names and **SHALL NOT** be reused for a different meaning. [\[CONF-HPDS-080\]](#)

15.2 Stability across versions

An element that retains its meaning in a later version **SHALL** retain its identifier. An element whose meaning changes materially **SHALL** receive a new identifier, and the change **SHALL** be recorded in the change record. [\[CONF-HPDS-081\]](#)

15.3 Version identifiers

Version identifiers **SHOULD** use the form major.minor.patch. The major number **SHOULD** be incremented for changes to clinical or behavioural meaning, the minor number for compatible additions, and the patch number for corrections that affect neither meaning nor structure.

15.4 References between descriptions

A reference to an element in another pathway description **SHALL** identify that description and its version, or an explicit permitted version range. [\[CONF-HPDS-082\]](#)

16 Authority and intended use

16.1 Authority status

Every pathway definition and implementation **SHALL** declare one authority status from Table 54. The additional requirements of Table 54 apply to the current authority status and are evaluated over the approvals and clinical authority recorded in the content and in the governance events of the same description (14.6). For a PathwayImplementation they are evaluated over its own attributes and events; approvals of the realised definition do not satisfy them (8.27). [\[CONF-HPDS-083\]](#)

Table 54: Authority status

Value	Meaning	Additional requirements
unapproved	No formal professional or organisational approval is asserted.	None.
reviewed	Content has undergone a declared review but is not represented as clinically approved.	ApprovalRecord with approvalType review.
clinically-approved	Clinical content has been approved by a recorded clinical or professional authority.	clinicalAuthority and ApprovalRecord with approvalType clinical.
operationally-authoritative	Approved by the owner for use in organising or governing care, in addition to clinical approval.	As clinically-approved, plus ApprovalRecord with approvalType organisational.

16.2 Intended use

A pathway description **SHALL** declare one or more intended uses from Table 55. Intended use is independent of authority status, but the combinations in the third column are required. [\[CONF-HPDS-084\]](#)

Table 55: Intended use

Value	Meaning	Minimum authority status
education	Explanation, teaching or illustration.	unapproved
analysis	Analysis, comparison, modelling or research.	unapproved
service-design	Designing or redesigning services and pathways.	unapproved
commissioning	Planning, purchasing or contracting services.	reviewed
implementation-specification	Specifying requirements for systems or local implementation.	reviewed
care-guidance	Guiding clinicians and staff in delivering care.	clinically-approved
care-organisation	Organising or governing the delivery of care.	operationally-authoritative
other	Another declared use.	unapproved

For the purpose of the third column, authority statuses are ordered as listed in Table 54, from unapproved to operationally-authoritative. Consumers **SHALL** display authority status and intended use wherever reliance on the pathway matters. [\[CONF-HPDS-085\]](#)

16.3 Automated use

This document does not define an executable authority status. Where pathway content is used to configure or drive automated clinical or operational activity, the pathway implementation **SHALL** identify each such artefact as an ImplementationArtefact with drivesAutomatedActivity true and **SHALL** reference the applicable assurance evidence. Clinical risk management obligations arise from the deployment context and are outside the scope of this document. [\[CONF-HPDS-086\]](#)

NOTE Applicable requirements depend on jurisdiction. For example, in England DCB0129 and DCB0160 apply to manufacturers and deploying organisations of health IT systems [14]. Software that drives clinical activity can fall within medical device regulation, for which ISO 14971 describes risk management [15, 16].

17 Human-readable catalogue representation

Every pathway definition **SHALL** be capable of presentation as a human-readable catalogue record. Every presentation of a pathway definition, in any view, **SHALL** display its name, identifier, version, current lifecycle status, current authority status, intended use and owner, and **SHALL** identify the source description and version from which it is produced. The remaining items of the minimum pathway definition dataset (5.3) **SHALL** be shown in the presentation or be available from it by direct navigation. Other content **SHOULD** be presented as applicable. [\[CONF-HPDS-087\]](#)

Appendix G gives a reference presentation that **SHOULD** be used for full catalogue records. Tabular, document and application views **MAY** order and group content differently.

Tabular and graphical views, including diagrams generated from formal representations, are projections of the underlying pathway definition. They **SHALL** identify the source pathway identifier and version and **SHALL NOT** become independent authoritative copies. [\[CONF-HPDS-088\]](#)

18 Machine-readable representation

At Level 2 and above, a pathway definition **SHALL** be transformable without semantic loss into a directed labelled graph in which elements are nodes and relationships are edges labelled with their type. Structural, behavioural, dependency, responsibility, context, outcome and derivation edges **SHALL** remain distinguishable by the categories in Table 47. [\[CONF-HPDS-089\]](#)

NOTE The graph supports comparison, dependency analysis, impact assessment, derivation of interoperability requirements and visualisation.

19 Serialisation

19.1 Normative JSON serialisation

The normative serialisation is JSON conforming to ISO/IEC 21778:2017, encoded as UTF-8, and valid against the JSON Schema with identifier urn:hpds:schema:0.5 (Appendix B).

NOTE The schema is provided in the WD 0.5 release package. It is generated from the attribute tables of this document and has been tested only against the examples and test cases reported in Appendix I.

19.2 Other serialisations

Other serialisations, such as JSON-LD [11], **MAY** be used where they preserve all information and identifiers and can be transformed to and from the normative serialisation without loss. A mapping that cannot round-trip without material loss **SHALL NOT** be represented as normative content. [\[CONF-HPDS-090\]](#)

19.3 Authority of the conceptual model

Where a serialisation, schema or mapping conflicts with Chapters 7 to 16, the conceptual model prevails and the inconsistency **SHALL** be treated as a defect in the serialisation, schema or mapping. Differences between a pathway definition and a formal representation are not handled by this rule; they are classified and handled as specified in 8.10.1 to 8.10.3. [\[CONF-HPDS-091\]](#)

Appendix A Validation rules

(normative)

A.1 General

Pathway description producers **SHALL** apply the rules in Table A.1. A rule of severity Error prevents a conformance claim for the levels or classes shown. A rule of severity Warning indicates a condition that requires human review; it **SHALL** be reported but does not prevent conformance. [CONF-HPDS-092]

Rule identifiers are persistent. A withdrawn rule retains its identifier, which is not reused; new rules are appended. The Status column shows whether a rule is unchanged from WD 0.4, revised, new in WD 0.5 or withdrawn.

The Method column states how a rule is evaluated: Structural (within one description, by schema or simple checks); Conditional (attribute conditions within one description); Graph (analysis of the behavioural graph, Chapter 9); Cross-version or Cross-description (requires another version or another referenced description); External (requires a terminology or publication service); Profile (requires the expression profile); Review (requires human judgement). Rules whose method is Review cannot be decided by software alone; a conformance claim relying on them **SHALL** state who performed the review. [CONF-HPDS-093]

So that independent validators report the same result, the following reporting conventions apply. A violation of the schema (Appendix B) is reported under V59; V02, V46, V52 and V53 are also reported when the items they name are absent. Where a rule other than V17 states a cardinality that is also stated in Table 47 (V18, V20, V21, V22, V26, V27, V33 and V46), a violation is reported under that rule and not under V17. A computable expression without an expression profile is reported under V42 and, where Table 52 requires a computable expression, also under V44.

NOTE In the Applies column, numbers are pathway definition conformance levels. Implementation and Catalogue denote the corresponding conformance classes.

Table A.1: Validation rules

Rule	Condition	Severity	Applies	Method	Source	Status
V01	Every element has an identifier that is unique within the description.	Error	1 to 4, Implementation, Catalogue	Structural	8.3; 15.1	unchanged
V02	The minimum pathway definition dataset (5.3) is present.	Error	1 to 4	Structural	5.3	unchanged
V03	At least one Population, EntryPoint, ExitPoint and Outcome is present, and no Outcome is defined solely as completion of an activity or stage.	Error	1 to 4	Structural; Review	5.3; 8.5; 8.6; 8.7; 8.20	unchanged
V04	Authority status and intended use are declared using values from Table 54 and Table 55.	Error	1 to 4, Implementation	Structural	16.1; 16.2	unchanged
V05	A description in the published state or later has effective and review dates.	Error	1 to 4, Implementation	Conditional	8.4; 14.1	unchanged
V06	The approval and clinical authority requirements of Table 54 are met for the current authority status of a definition or implementation, evaluated over the approvals and clinical authority recorded in its own content and governance events.	Error	1 to 4, Implementation	Conditional	16.1; 8.27; 14.2; 14.6	revised
V07	Every declared intended use has at least the minimum authority status in Table 55.	Error	1 to 4, Implementation	Conditional	16.2	unchanged
V08	A pathway variant identifies its parent and parent version.	Error	1 to 4	Conditional	8.4; 13.1	unchanged
V09	No care-model element of a pathway definition, and no formal representation it references, identifies an implementation-specific delivery arrangement as defined in 6.1. Named organisations and people in governance and provenance metadata are permitted.	Error	1 to 4	Review	6.1; 8.10	revised
V10	A definition with no DecisionPoint has noDecisionPoints set to true.	Error	1 to 4	Conditional	8.4	unchanged
V11	The effective date is earlier than the current review date.	Error	1 to 4, Implementation	Conditional	8.4; 14.1	revised
V12	The current review date has passed while the description remains published.	Warning	1 to 4, Implementation	Conditional	14.1; 14.6	revised
V13	Every modifying extension used is listed in modifyingExtension.	Error	1 to 4, Implementation	Structural	13.6	unchanged

Rule	Condition	Severity	Applies	Method	Source	Status
V14	The content of a published version is identical in every retrieved copy; governance-state attributes change only by appended governance events or discrepancy records, and existing governance events are not altered or removed.	Error	1 to 4, Implementation, Catalogue	Cross-version	14.2	revised
V15	Every version after the first published version has a ChangeRecord.	Error	1 to 4, Implementation	Cross-version	14.4	unchanged
V16	Every reference resolves to an element or identified structured value in the description, or to a version-qualified element in another description. References to external resources listed in 8.2 are exempt.	Error	2 to 4, Implementation	Structural; Cross-description	8.2; 15.4	revised
V17	Every relationship has a permitted source and target type and does not exceed the maximum cardinalities in Table 47.	Error	1 to 4	Graph	9.1; 9.2	revised
V18	Every PathwayStage, Activity and DecisionPoint, and every Event whose external attribute is not true, has exactly one structural container; no Event whose external attribute is true has a container.	Error	2 to 4	Graph	9.3	revised
V19	Withdrawn in WD 0.4 with the Task entity. The identifier is not reused.	–	–	–	–	withdrawn
V20	Every EntryPoint has exactly one startsAt relationship.	Error	2 to 4	Graph	8.6; 9.2	unchanged
V21	Every DecisionPoint selects at least two PathwayBranch elements, and each branch belongs to exactly one DecisionPoint.	Error	2 to 4	Graph	8.11; 9.2	unchanged
V22	Every PathwayBranch has exactly one branchTo relationship and at most one rejoins relationship.	Error	2 to 4	Graph	8.12; 9.2	unchanged
V23	Every branch of an exclusive or inclusive DecisionPoint has a condition or is its single default branch.	Error	2 to 4	Graph	8.12	unchanged
V24	Join rules are present and consistent where required by Table 46, and joinCount is present where joinRule is count.	Error	2 to 4	Graph	9.1	revised
V25	Every Activity with requiresClinicalResponsibility true has at least one performs relationship.	Error	2 to 4	Graph	8.9	unchanged
V26	Every DecisionPoint has at least one decides relationship.	Error	2 to 4	Graph	8.11; 9.2	unchanged
V27	Every InformationRequirement states a purpose and is required by at least one element.	Error	2 to 4	Graph	8.17	unchanged
V28	Every behavioural node other than an Event whose external attribute is true is reachable as defined in 9.6.	Error	2 to 4	Graph	9.6	revised
V29	For every EntryPoint, at least one ExitPoint is reachable as defined in 9.6 f).	Error	2 to 4	Graph	9.6	revised
V30	The cycle-detection graph defined in 9.5 is acyclic.	Error	2 to 4	Graph	9.5	revised
V31	An ExitPoint with exitType transfer has a transitionsTo relationship, and an EntryPoint with entryType transition has fromPathway.	Error	2 to 4	Graph	8.6; 8.7	unchanged
V32	Every includes, transitionsTo, specialises, realises and cross-description escalatesTo target is version qualified.	Error	1 to 4, Implementation	Structural	9.2; 15.4	revised
V33	Every Measure assesses at least one element.	Error	2 to 4	Graph	8.21	unchanged
V34	An information product of an element provided by one service is required by an element provided by another service without a covering InteroperabilityRequirement.	Warning	2 to 4	Graph	Chapter 12	unchanged
V35	Every temporal constraint that requires an anchor has one, and the anchor resolves to an element.	Error	2 to 4	Structural	10.1	unchanged
V36	Temporal constraints applying to the same elements cannot be satisfied together.	Warning	2 to 4	Review	10.3	unchanged
V37	Every Deviation other than an addition identifies a parent element that exists in the referenced parent version.	Error	2 to 4, Implementation	Cross-version	13.2	unchanged
V38	Every terminology binding identifies resource, version and code or data element, and every value set binding states its strength.	Error	3 to 4	Structural	11.2	unchanged
V39	At Level 3 and above, the Expressions of population, entry and exit criteria identify the clinical concepts they depend on, and every ClinicalConcept so referenced, or referenced by a mandatory InformationRequirement, is bound or declared local.	Error	3 to 4	Structural	11.3	revised
V40	A bound resource version has been withdrawn or superseded by its publisher.	Warning	3 to 4	External	11.4	unchanged
V41	Every change of bound resource version is recorded in a ChangeRecord.	Error	3 to 4	Cross-version	11.4	unchanged
V42	Every computable expression present, at any level, identifies an expression profile, language and language version.	Error	1 to 4	Structural	8.26.2; 13.4	revised
V43	Every expression profile used identifies its evaluation context and the other content required by 13.4.	Error	4	Profile	13.4	unchanged

Rule	Condition	Severity	Applies	Method	Source	Status
V44	Every attribute designated Mandatory at Level 4 in Table 52 carries a computable expression in every occurrence, subject to the exceptions stated there.	Error	4	Structural	13.7	revised
V45	Every DecisionPoint that is not judgement-based has logic provenance.	Error	4	Structural	8.11	unchanged
V46	A PathwayImplementation identifies the definition and version it realises.	Error	Implementation	Structural	5.4; 8.27	unchanged
V47	Every difference from the realised definition is recorded as a Deviation with a rationale.	Error	Implementation	Review	13.2; 8.27	unchanged
V48	Every mapping identifies an element that exists in the realised definition version.	Error	Implementation	Cross-description	5.4; 8.28.1	unchanged
V49	Every ImplementationArtefact with drivesAutomatedActivity true references assurance evidence.	Error	Implementation	Conditional	16.3; 8.28.2	unchanged
V50	Every catalogue entry resolves to a pathway definition identifier and version.	Error	Catalogue	Cross-description	5.5	unchanged
V51	Catalogue entry identifiers are unique within the catalogue and unchanged from earlier catalogue releases.	Error	Catalogue	Cross-version	5.5	unchanged
V52	Catalogue publisher, governance authority, scope and version are present.	Error	Catalogue	Structural	5.5; 7.1	unchanged
V53	Every FormalRepresentation identifies its representation type, standard, locator, representation version and correspondence.	Error	1 to 4	Structural	8.10	unchanged
V54	No FormalRepresentation referenced by a pathway definition contains implementation-specific content (6.1, 8.10).	Error	1 to 4	Review	6.1; 8.10	unchanged
V55	No transitionsTo or escalatesTo relationship has a HealthcarePathway as its source.	Error	2 to 4	Graph	9.2	unchanged
V56	Every PathwayStage with ordering unordered has at least one completion condition.	Error	1 to 4	Conditional	8.8	unchanged
V57	A pathway definition in the published state or later has no rights statement.	Warning	1 to 4	Conditional	14.1	unchanged
V58	Every localisesRepresentation reference resolves to a FormalRepresentation of the realised definition version.	Error	Implementation	Cross-description	8.28.2	unchanged
V59	Every attribute with obligation M is present, every attribute with obligation C is present where its condition holds, and every attribute respects its multiplicity.	Error	1 to 4, Implementation, Catalogue	Structural; Conditional	8.1; 1.8.3	new
V60	At Level 2 and above, the source of every includes relationship is an Activity.	Error	2 to 4	Graph	9.2; 9.3	new
V61	The deviations of a pathway variant are complete as defined in 13.2 when compared with the referenced parent version.	Error	2 to 4	Cross-version	13.2	new
V62	Governance events are in non-decreasing date order, and lifecycleStatus, authorityStatus and reviewDate each equal the value established by the latest governance event that sets them.	Error	1 to 4, Implementation	Conditional	14.2; 14.6	new
V63	A description whose lifecycle status is published or any later state has a governance event of type publication.	Error	1 to 4, Implementation	Conditional	14.2	new
V64	An ordered stage has more than one initial node.	Warning	2 to 4	Graph	8.8; 9.6	new
V65	A reachable behavioural node directly contained in the HealthcarePathway has no route to an ExitPoint.	Warning	2 to 4	Graph	9.6; 9.9	new
V66	A formal representation has a discrepancy of type contradiction or unresolved whose status is open.	Warning	1 to 4	Conditional	8.10.1; 8.10.3	new
V67	The target of every waitsFor relationship is an Event whose external attribute is true or a reachable Event.	Error	2 to 4	Graph	9.6	new
V68	The source of every iterates relationship is reachable from its target in the cycle-detection graph.	Error	2 to 4	Graph	9.5	new

Appendix B JSON serialisation

(normative)

B.1 General

The normative schema for this working draft has the identifier urn:hpds:schema:0.5. It supports three document forms, identified by the `documentType` property, as shown in Table B.1.

Table B.1: Document forms

documentType	Content
definition	One HealthcarePathway, with its formal representation references, deviations and governance events, plus the element collections in Table B.2.
implementation	One PathwayImplementation plus any local elements and relationships needed to represent implementation-specific content.
catalogue	One PathwayCatalogue manifest referencing pathway definitions by identifier and version.

NOTE The schema file `hpds.schema.0.5.json` is provided in the WD 0.5 release package. It is generated from the attribute tables of this document. It has been tested only against the examples and test cases listed in Appendix I, using one JSON Schema implementation. It has not been reviewed independently (Appendix N).

B.2 Serialisation rules

The serialisation **SHALL** meet the following requirements: [\[CONF-HPDS-094\]](#)

- property names correspond to the attribute and component names in Chapters 7 to 10;
- element collections are held in the top-level arrays named in Table B.2, and relationships are held in the `relationships` array and refer to elements by identifier;
- a `Reference` is serialised as an object with the components `ref`, `pathwayDescription`, `version` and `versionRange` (8.2), and a `CodeableText` as an object with the components `text` and `binding`, where `binding` holds zero or more `TerminologyBinding` values;
- formal representations are referenced by `locator` and are not embedded;
- governance-state attributes (14.2) are serialised with the content of the element that holds them; they are not part of the immutable content;
- undeclared properties are not permitted; additions **SHALL** use the extension mechanism (13.6); [\[CONF-HPDS-095\]](#)
- enumerated values are validated against the value sets of this document; restrictions applied by domain profiles are validated by rules defined in those profiles.

The schema enforces structure, value sets, mandatory attributes and multiplicities. Conditional obligations, graph, cross-version and semantic rules in Appendix A require additional validation.

B.3 Collections

Table B.2: Element collections in a definition document

Property	Content
<code>pathway</code>	The <code>HealthcarePathway</code> element (exactly one).
<code>populations</code>	Population elements.
<code>entryPoints</code>	<code>EntryPoint</code> elements.
<code>exitPoints</code>	<code>ExitPoint</code> elements.
<code>stages</code>	<code>PathwayStage</code> elements.
<code>activities</code>	Activity elements.
<code>decisionPoints</code>	<code>DecisionPoint</code> elements.
<code>branches</code>	<code>PathwayBranch</code> elements.
<code>events</code>	Event elements.
<code>roles</code>	Role elements.
<code>services</code>	Service elements.
<code>careSettings</code>	<code>CareSetting</code> elements.

Property	Content
informationRequirements	InformationRequirement elements.
informationProducts	InformationProduct elements.
clinicalConcepts	ClinicalConcept elements.
outcomes	Outcome elements.
measures	Measure elements.
constraints	Constraint elements.
exceptions	Exception elements.
evidenceSources	EvidenceSource elements.
interoperabilityRequirements	InteroperabilityRequirement elements.
relationships	Relationship elements.

An implementation document holds its PathwayImplementation in the property implementation and **MAY** hold local elements in the same collections. A catalogue document holds its PathwayCatalogue in the property catalogue.

Appendix C Mapping to HL7 FHIR

(informative)

FHIR PlanDefinition is a possible formal representation of aspects of a pathway definition and a possible implementation artefact [7, 8]. This document does not require FHIR, and a conformant pathway definition is not expected to be representable without loss in FHIR resources without profiles or extensions. Table C.1 identifies principal correspondences. The mapping should be confirmed against the FHIR release and implementation guides in use.

Table C.1: Correspondence with FHIR resources

HPDS concept	FHIR correspondence	Observations
HealthcarePathway	PlanDefinition	FHIR represents a shareable plan and actions; catalogue governance, authority status, intended use and explicit deviations require extensions or conventions.
FormalRepresentation	PlanDefinition, Library (CQL)	A technology-neutral PlanDefinition is a formal representation with representationType plan-definition; a localised one is an implementation artefact.
PathwayStage	Nested PlanDefinition.action	Stage semantics require convention. An unordered stage corresponds to actions without relatedAction ordering.
Activity	PlanDefinition.action; ActivityDefinition	Clinical responsibility and completion criteria may require profiling.
DecisionPoint, PathwayBranch	Action with condition and selection behaviour; CQL libraries	Judgement-based and shared decisions, uncertainty and branch rejoin require extensions or conventions.
EntryPoint	Action trigger or applicability condition	Entry routes and criteria are catalogue concepts in HPDS.
Event	Action trigger definitions	Interruption and escalation effects require convention.
precedes	Action relatedAction	Join rules are not directly represented.
includes	Action definition referencing another PlanDefinition	Version qualification requires care.
accountableFor, performs, decides	Action participant	Accountability as distinct from performance requires convention.
InformationRequirement	Action input (DataRequirement); Questionnaire	Purpose and quality requirements require extension.
InformationProduct	Action output	Instance resources are outside HPDS.
Outcome	PlanDefinition.goal	HPDS keeps intended outcome distinct from process completion.
Measure	Measure	Direct correspondence for measure definition.
EvidenceSource	RelatedArtifact	Direct correspondence for source linkage.
Exception	No direct correspondence	Requires extension or modelling as conditional actions.
Deviation	No direct correspondence	Requires extension, related artefact or external governance record.
PathwayImplementation, ImplementationArtefact	No single resource	Artefacts may themselves be FHIR resources referenced by locator.
Pathway instance reference points (8.29)	CarePlan, Task, ServiceRequest, Encounter, Observation, Procedure and related resources	Outside the normative scope; related through instantiation references.

NOTE The HL7 FHIR Clinical Practice Guidelines implementation guide [8] describes graded levels of knowledge representation from narrative to executable. Their correspondence with the conformance levels in 5.2 is to be confirmed (Appendix N); in particular, no HPDS level corresponds to executable content.

Appendix D Mapping to BPMN, CMMN and DMN and relationship with the HL7 Field Guide

(informative)

D.1 General

BPMN, CMMN and DMN provide precise views of process, adaptive case behaviour and decision logic [2, 3, 4]. These views can be derived from, or aligned with, a pathway definition and referenced as formal representations (8.10) where greater behavioural or decision precision is required, or used in implementation artefacts (8.28.2) once localised. This document does not require any one notation. A model acquires executable status only through the additional design and assurance described in 9.7 and 16.3.

D.2 Correspondence with BPMN, CMMN and DMN

Table D.1: Correspondence with BPMN, CMMN and DMN

HPDS concept	Possible representation	Observations
HealthcarePathway	BPMN process or CMMN case	Choice depends on how prescriptive the behaviour is.
PathwayStage (ordered); includes	BPMN sub-process or call activity; CMMN stage	
PathwayStage (unordered)	CMMN stage with discretionary items; BPMN ad hoc sub-process	HPDS does not adopt CMMN sentry or plan-item lifecycle semantics.
Activity	BPMN task; CMMN task	Activity.optional corresponds to a CMMN discretionary item.
DecisionPoint (exclusive)	BPMN exclusive gateway preceded by a business rule task	Criteria may be expressed as DMN decisions.
DecisionPoint (inclusive)	BPMN inclusive gateway	
DecisionPoint (deferred)	BPMN event-based gateway	
Several precedes from one source; joinRule	BPMN parallel, inclusive or complex gateway	A conditional precedes corresponds to a conditional sequence flow.
Event; EntryPoint; ExitPoint	BPMN start, intermediate and end events; CMMN event listener	
Temporal constraint	BPMN timer event; CMMN timer event listener	Anchored constraints only.
interrupts; escalatesTo	BPMN boundary and escalation events	
Exception	BPMN boundary event or conditional flow; CMMN discretionary item	
Role, Service	BPMN lane; CMMN case role	Lanes should represent roles or services, not named organisations.
accountableFor	Accountable assignment in a responsibility matrix	Not represented natively in BPMN or CMMN diagrams.
InteroperabilityRequirement	BPMN message flow in a collaboration view	Pools representing named organisations are appropriate only in implementation views.
InformationRequirement, InformationProduct	BPMN data input, data output and data object; CMMN case file item; DMN input data	Purpose and bindings are not represented natively.
EvidenceSource supporting decision logic	DMN knowledge source	Direct correspondence for decision authority.

D.3 Relationship with the HL7 Field Guide to Shareable Clinical Pathways

HL7 Guidance: Field Guide to Shareable Clinical Pathways, Edition 3 [5] is an HL7 informative document that gives a method and style for producing, reading and consuming clinical pathway models expressed in BPMN, CMMN and DMN. It treats a shareable pathway as a set of linked models, organised from a structured library of activities, actors, data, events, resources and goals, and intended to be refined iteratively until the models are executable. It distinguishes a platform-independent shared model from the platform-specific model that a receiving organisation produces through localisation.

The two documents address different layers and are complementary. Table D.2 summarises the difference.

Table D.2: Positioning of this document and the HL7 Field Guide

Aspect	This document	HL7 Field Guide
Nature	Specification of a governed business artefact, with conformance requirements	Informative method and style guidance for modelling
Primary unit	Pathway definition held in a pathway catalogue	Set of linked BPMN, CMMN and DMN models
Intended formality	Business meaning; computable expressions optional at Level 4; never executable	Formal models intended to become executable
Principal audience	Clinicians, managers, architects and catalogue governors	Modellers and knowledge engineers, with clinical review
Governance	Normative identity, versioning, lifecycle, authority, intended use, provenance and deviation	Informative guidance on asset metadata, curation and ecosystems
Localisation	Pathway implementation with explicit mappings and deviations	Platform-specific model produced by the consumer
Technology neutrality	Required of definitions and formal representations	Aimed for in shared models; localised models are platform specific

Table D.3: Correspondence between Field Guide constructs and this document

Field Guide construct	Correspondence in this document
Shareable clinical pathway (platform-independent model set)	One or more FormalRepresentations of a pathway definition, with correspondence derived, source or aligned (8.10).
Localised (platform-specific) model	ImplementationArtefact of a PathwayImplementation, with localisesRepresentation (8.28.2).
Library model: activities	Activity and PathwayStage.
Library model: actors and responsibility assignments	Role with performs, decides and accountableFor relationships; Service. Organisational participants belong in implementations.
Library model: data items (situational data)	InformationRequirement, InformationProduct and ClinicalConcept, with terminology bindings.
Library model: events	Event, EntryPoint and ExitPoint.
Library model: systems (non-person resources)	Not part of a pathway definition. Recorded as ImplementationMapping or ImplementationArtefact.
Library model: goals	Outcome.
Knowledge model diagram	Graph view (Chapter 18) including includes, transitionsTo and escalatesTo relationships between pathways.
Decision services and knowledge sources	DecisionPoint with logicProvenance and EvidenceSource; DMN models as formal representations.
Fixed decision outputs shared across localisations	Branches of a DecisionPoint define permitted outcomes; changes are Deviations (8.11, 13.2).
Asset metadata for discovery and curation	HealthcarePathway governance attributes, CatalogueEntry and rights (8.4, 7.2).
Variance record	Instance level; supported by reference points (8.29).

D.4 Decision

This document remains independent of the HL7 Field Guide. It references models produced under the Field Guide as formal representations or implementation artefacts, maps its concepts to them as in Table D.3, and does not incorporate the Field Guide's modelling method, notational conventions or execution orientation. The rationale is recorded in K.5.

Appendix E Mapping to ArchiMate

(informative)

Table E.1 relates pathway content to ArchiMate elements [6] so that pathway definitions and implementations can be connected to enterprise architecture models. ArchiMate is not required for pathway description.

Table E.1: Correspondence with ArchiMate

HPDS concept	ArchiMate correspondence	Observations
HealthcarePathway	Business process or value stream	Choice depends on modelling purpose.
PathwayStage, Activity	Business process or business function	
Event	Business event	
Role	Business role	
Service	Business service	Providers are business actors in implementation views.
CareSetting	Location or business interaction context	
InformationRequirement, InformationProduct	Business object; data object in implementation views	
Outcome; purpose	Outcome; goal	
Measure	Assessment or property of an outcome	No dedicated measure element.
Constraint	Constraint	
InteroperabilityRequirement	Requirement and flow relationship	
ImplementationMapping, ImplementationArtefact	Business actor, application component, application service, technology service, artefact	
Capability exercised by a pathway	Capability	Derived by analysis rather than recorded in the pathway definition (K.5).

Appendix F Relationship with ISO 13940:2026 and the standards landscape

(informative)

F.1 ISO 13940:2026

ISO 13940 specifies requirements for a system of concepts to define care processes that support continuity of care, and applies to social care as well as clinical care. This document uses ISO 13940:2026 as its principal conceptual reference and describes particular pathway business artefacts rather than reproducing the ISO 13940 concept system. The 2026 edition is understood from a national adoption record to replace ISO 13940:2015 [17]; its publication and final text have not been verified for this working draft. Before any proposal for formal standardisation, the terms in Chapter 3 are to be reconciled concept by concept with the published ISO 13940:2026 text; conflicting definitions are to be removed or stated as specialisations.

F.2 WHO SMART Guidelines and Digital Adaptation Kits

WHO Digital Adaptation Kits are software-neutral structured documentation packages that can include health interventions, personas, user scenarios, workflows, core data elements, decision support logic, indicators and functional requirements [9]. This document is narrower. It provides the governed pathway catalogue artefact and can reference DAK content as a formal representation with representationType guideline-package, or map to it, without replacing the wider package.

F.3 Standards landscape

Table F.1: Position of this document relative to other standards

Standard or approach	Primary concern	Relationship with this document
ISO 13940:2026	System of concepts for continuity of care	Conceptual reference and terminology alignment
ISO 13606 [1]	Electronic health record communication	Instance level; outside normative scope
SNOMED CT [10], ICD and national terminologies	Terminology and classification	Targets of semantic binding
National data dictionaries and record standards, such as PRSB standards in the United Kingdom	Data definitions and record content	Binding targets for information requirements and products
HL7 FHIR and Clinical Practice Guidelines [7, 8]	Exchange, plan definitions and computable guideline artefacts	Formal representations and implementation artefacts (Appendix C)
BPMN, CMMN and DMN [2, 3, 4]	Process, case and decision modelling	Formal representations and implementation artefacts (Appendix D)
HL7 Field Guide to Shareable Clinical Pathways [5]	Method and style for BPMN, CMMN and DMN pathway models	Complementary modelling method; its outputs are referenced, not incorporated (D.3, D.4)
WHO SMART Guidelines and DAKs [9]	Software-neutral guideline adaptation packages	Potential source and formal representation
ArchiMate [6]	Enterprise architecture	Architecture views and implementation mapping (Appendix E)
UML	General-purpose modelling	May express the conceptual model

Appendix G Human-readable pathway catalogue record

(normative)

Table G.1 gives the reference presentation of a full catalogue record. It **SHOULD** be used for full catalogue records and document views. Other views **MAY** order and group content differently, provided that they meet Chapter 17: the identification items of section 0 are displayed, the source description and version are identified, and the remaining minimum dataset items are shown or directly available. In the reference presentation, a section for which the description contains no content **SHOULD** state that none is recorded.

Table G.1: Catalogue record sections

Section	Content
0 Identification	Name, identifier, version, current lifecycle status, current authority status, intended use, conformance level, owner, clinical authority, effective and current review dates, and any open representation discrepancy or unsupported modifying extension.
1 Purpose and scope	Purpose, scope and pathway classification.
2 Population	Inclusion, exclusion and stratification criteria.
3 Entry	Entry points, routes and entry criteria.
4 Outcomes and exit	Intended outcomes, exit points and exit criteria.
5 Pathway structure	Stages, their purposes and ordering, and included sub-pathways.
6 Activities and decisions	Activities, decision points, branches, events and principal behavioural relationships.
7 Roles, services and settings	Roles and services associated with activities and decisions, coordination accountability, and care settings.
8 Information dependencies	Information requirements, products, information-governance constraints and interoperability requirements.
9 Related pathways	Transitions, escalations and inclusions to and from other pathways.
10 Exceptions and variation	Exceptions and, where relevant, parent specialisation and deviations.
11 Measures	Measures and the elements they assess.
12 Evidence	Evidence sources and supported elements.
13 Governance, rights and provenance	Approvals, governance events, rights, provenance, profiles, formal representations and their discrepancies, and change history.
14 Implementations	Known pathway implementations, where the catalogue exposes them.

Appendix H Mapping existing tabular catalogues

(informative)

H.1 Method

Existing pathway catalogues can be imported by classifying each field as an entity, an attribute, a relationship, governance or provenance information, implementation-specific information, or a compound field that requires decomposition. The mapping should:

- record, for each source field, the target entity and attribute or relationship;
- identify fields that combine several concepts and decompose them;
- separate definition content from implementation content;
- record provenance with origin imported-from-catalogue (14.3);
- record information that cannot be represented without loss as a candidate model defect or profile requirement;
- confirm that the resulting pathway definitions remain recognisable business artefacts rather than procedure lists or system inventories.

NOTE A catalogue column structure is a view of pathway information. Reproducing it as a metamodel carries any combination of concepts in its columns into the model.

H.2 Findings from the Frailty and Multimorbidity Neighbourhood Care Pathway Catalogue

The Frailty and Multimorbidity Neighbourhood Care Pathway Catalogue (version 0.1.0, 38 entries in nine coordination domains) was mapped to WD 0.3 and used to test this revision. Its pathways are cross-organisational, largely adaptive and heavily interrelated. Table H.1 records the findings that led to changes or decisions. Entries in the catalogue are unapproved drafts for local validation; the findings concern the model, not the clinical content.

Table H.1: Catalogue test findings

Catalogue field or feature	Finding against WD 0.3	Response in WD 0.4
Care Coordinator / Accountable Function	No construct for accountability distinct from performance; recorded as a pending extension.	accountableFor relationship (9.2).
Related Pathways (includes, transitionsTo, escalatesTo)	Relationships recorded between whole pathways; WD 0.3 required an ExitPoint or Event source not yet authored.	Pathway-level sources permitted at Level 1, refined at Level 2 (9.2 NOTE 3, V55).
Principal Activities in pathways that are explicitly non-linear	Reachability rules required precedes relationships that do not exist in practice.	Unordered stages (8.8, 9.6).
Outstanding Level 1 items: principal stages	Stages absent for most entries and artificial when imposed.	Principal stages recommended rather than required (5.3).
Outstanding Level 1 items: owner, exit points, governance authority	Genuinely missing information needed for comparison and governance.	Requirements retained.
Required Digital and Care Coordination Capabilities	Mixed restated activities with implementation terms such as worklists.	No capability construct; capabilities remain derived (Appendix E, K.5).
Information Sharing / Consent / Access Requirements	Mapped to regulatory constraints without a precise type.	Constraint type information-governance; accessRequirement on InteroperabilityRequirement.
Shared Care Plan / Coordination Requirements	Mapped adequately to InformationProduct and Constraint.	NOTE added to 8.18; no model change.
Artefact Class: governance process	Entries whose subject is a service, not a subject of care.	Outside scope (1.2); not catalogue entries (7.1 NOTE).
Evidence Status	Mapped to EvidenceSource.strength and ProvenanceRecord.note.	ProvenanceRecord.note clarified; no model change.
Local Validation Questions	Implementation authoring aid, not definition content.	Confirmed as outside the definition; no model change.

Appendix I Examples, test cases and results

(informative)

I.1 Status of worked examples

Worked examples are drawn from identified sources, serialised in accordance with Appendix B, and declared with authority status unapproved and intended use education or analysis. They contain no clinical recommendations beyond those stated in their sources and imply no clinical approval. Table I.1 states which examples have been produced for the WD 0.5 release package and which remain outstanding.

Table I.1: Worked examples

Example	Status	Source	Characteristics demonstrated
Hello Patient visit (sequential)	Produced, Level 2	HL7 Field Guide to Shareable Clinical Pathways, Edition 3, sections 3 and 4 and Appendix D (illustrative models) [5]	Sequential flow; ordered and unordered stages; exclusive decision with DMN logic provenance; formal representations with correspondence source; named owner in governance metadata.
Comprehensive geriatric assessment (adaptive)	Produced, Level 2	Frailty and Multimorbidity Neighbourhood Care Pathway Catalogue v0.1.0, entry FNM-CGA-001, and its source register	Unordered stage; sub-pathways included from activities; judgement-based inclusive decision; transfers and escalation to other pathways; coordination accountability; information-governance constraint.
Two-hour urgent community response (catalogue record)	Produced, Level 1	Same catalogue, entry FNM-UCR-002	Minimal catalogue record; pathway-level relationships; no behavioural graph.
Catalogue manifest	Produced	Same catalogue	Catalogue entries referencing the three definitions by identifier and version.
Hello Patient implementation	Produced	Derived from the sequential example	Implementation with its own authority; localised artefact referencing a formal representation. Implementing organisation is fictional.
Endoscopy, maternity, pathology, acute admission and discharge	Outstanding	To be identified	As listed in WD 0.4 Annex I.
WHO SMART guideline or DAK mapping	Outstanding	To be identified	Independent mapping test.

I.2 Test cases and results

Each test case is a complete description, or a pair of descriptions for cross-version rules, with its expected result: the declared conformance level or class, and the set of Error and Warning rules expected to fail. Invalid test cases are derived from the valid examples by a single deliberate change. Table I.2 lists the test cases and the result obtained from the reference validator in the release package. A result of Pass means that the rules reported by the validator were exactly the expected rules.

Table I.2: Test cases and results

Test	Description	Finding	Expected	Reference validator	Result
T01	Sequential example, Level 2 (valid)	2, 3	No errors or warnings	No errors or warnings	Pass
T02	Adaptive example, Level 2 (valid)	2	No errors or warnings	No errors or warnings	Pass
T03	Catalogue record, Level 1, with pathway-level relationships (valid)	9	No errors or warnings	No errors or warnings	Pass
T04	Catalogue manifest (valid)	–	No errors or warnings	No errors or warnings	Pass
T05	Implementation with its own authority (valid)	4	No errors or warnings	No errors or warnings	Pass
T06	Entry point without startsAt	2	V20, V28, V29	V20, V28, V29	Pass
T07	Cycle through a decision point and branch without iterates	2	V30	V30	Pass
T08	Repetition expressed by iterates with a repeated route (valid)	2	No errors or warnings	No errors or warnings	Pass
T09	iterates whose source is not reachable from its target	2	V68	V68	Pass

Test	Description	Finding	Expected	Reference validator	Result
T10	Unordered stage without completion condition	2	V56	V56	Pass
T11	Ordered stage with several initial nodes	2	V64 (warning)	V64 (warning)	Pass
T12	includes from the pathway at Level 2	2	V60	V60	Pass
T13	External anchor event without container or effect (valid)	2	No errors or warnings	No errors or warnings	Pass
T14	External event given a container	2	V18	V18	Pass
T15	Relationship with a source type not permitted, at Level 1	2, 11	V17	V17	Pass
T16	Reachable top-level node with no route to an exit	2	V30, V65 (warning)	V30, V65 (warning)	Pass
T17	Named organisations in governance metadata (valid)	3	No errors or warnings	No errors or warnings	Pass
T18	Published version with publication event (valid)	5	No errors or warnings	No errors or warnings	Pass
T19	Published version without publication event	5	V63	V63	Pass
T20	Later copy with appended lifecycle event (valid)	5	No errors or warnings	No errors or warnings	Pass
T21	Lifecycle status inconsistent with latest event	5	V62	V62	Pass
T22	Published content changed in place	5	V14	V14	Pass
T23	Clinically approved definition without clinical authority	4	V06	V06	Pass
T24	Clinically approved implementation without its own clinical authority	4	V06	V06	Pass
T25	Clinically approved implementation with its own clinical authority (valid)	4	No errors or warnings	No errors or warnings	Pass
T26	Variant with complete deviations (valid)	4	No errors or warnings	No errors or warnings	Pass
T27	Variant with an omitted element not recorded as a deviation	4	V61	V61	Pass
T28	Level 3 with criteria concepts declared local (valid; binding report)	4, 8	No errors or warnings	No errors or warnings	Pass
T29	Level 3 concept neither bound nor declared local	4, 8	V39	V39	Pass
T30	Level 3 criterion identifying no concept	4	V39	V39	Pass
T31	Level 4 with designated expressions computable under the test fixture (valid structurally)	6	No errors or warnings	No errors or warnings	Pass
T32	Level 4 with narrative-only branch conditions of a non-judgement decision	6	V44	V44	Pass
T33	Level 4 judgement-based decision: criteria and branch conditions may be narrative (valid)	6	No errors or warnings	No errors or warnings	Pass
T34	Computable expression without profile	6	V42, V44	V42, V44	Pass
T35	Modifying extension not listed	8	V13	V13	Pass
T36	Modifying extension listed (valid for a producer)	8	No errors or warnings	No errors or warnings	Pass
T37	Formal representation with an open unresolved discrepancy	7	V66 (warning)	V66 (warning)	Pass
T38	Resolved discrepancy without resolution	7, 11	V59	V59	Pass
T39	Formal representation without correspondence	7	V53, V59	V53, V59	Pass
T40	Missing owner	9, 11	V02, V59	V02, V59	Pass

NOTE The results show that the reference validator agrees with the expected results. They do not show that an independent implementation would reach the same results; that test is outstanding (Appendix N).

Appendix J Conformance checklist

(normative)

A conformance claim **SHALL** be supported by a completed checklist in the form of Table J.1. Items that do not apply to the conformance class claimed **SHALL** be marked not applicable. [\[CONF-HPDS-096\]](#)

Table J.1: Conformance checklist

Item	Requirement	Reference	Result
J1	Version of this document and conformance class stated.	5.1	
J2	Conformance level declared for a pathway definition.	5.2	
J3	Domain and expression profiles identified with versions.	5.1, 13.3 to 13.5	
J4	Minimum pathway definition dataset present.	5.3	
J5	No implementation-specific delivery arrangements in the care-model content of a definition or its formal representations; named organisations appear only in governance and provenance metadata.	6.1, 8.10	
J6	No personal health information present.	6.1	
J7	Required entities, attributes and conditional attributes present.	Chapters 7 and 8	
J8	Structural relationships conform to Chapter 9.	9.2, 9.3	
J9	Behavioural relationships, stage ordering, join rules, iteration, reachability and cycles conform to Chapter 9.	8.8, 9.1, 9.4 to 9.9	
J10	Temporal constraints conform to Chapter 10.	Chapter 10	
J11	Semantic bindings conform to Chapter 11.	Chapter 11	
J12	Interoperability requirements conform to Chapter 12.	Chapter 12	
J13	Variants and implementations reference parents and record deviations.	13.1, 13.2	
J14	Extensions declared; modifying extensions listed.	13.6	
J15	Governance, rights, provenance, change history and lifecycle recorded.	Chapter 14	
J16	Identifiers and versions conform to Chapter 15.	Chapter 15	
J17	Authority status and intended use recorded and consistent.	Chapter 16	
J18	Automation artefacts identified with assurance evidence.	8.28.2, 16.3	
J19	Human-readable catalogue record available.	Chapter 17, Appendix G	
J20	Serialisation validates against hpds.schema.0.5.json.	Chapter 19, Appendix B	
J21	No validation errors for the declared level or class; warnings reported.	Appendix A	
J22	Formal representations identified, technology neutral and not the sole location of required content; contradictions and unresolved discrepancies recorded.	8.10	
J23	Governance events recorded from publication; governance-state attributes consistent with them.	14.2, 14.6	
J24	For Level 3 or 4, binding report included.	11.5	
J25	Rules evaluated by Review identified, with the reviewer; release package version identified.	Appendix A, 5.1	

Appendix K Development, validation and publication

(informative)

K.1 Development priorities following WD 0.5

The prioritised list of unresolved issues and missing implementation evidence is given in Appendix N. It supersedes the development priorities and open issues listed in WD 0.4.

K.2 Repository structure

Table K.1: Repository structure

Location	Content
/specification	HTML and document versions of this document
/model	Authoritative metamodel source and generated model views, including the information model workbook
/schema	hpds.schema.0.5.json and earlier schemas
/examples	Worked examples
/profiles	Domain and expression profiles
/mappings	FHIR, BPMN, CMMN, DMN, HL7 Field Guide, ArchiMate and other mappings
/catalogue	Catalogue templates, manifests, migration records and test findings
/governance	Conformance checklist, decision register, issue process and change proposals

K.3 Success criteria

The development is to be judged successful if:

- substantially different pathways, including sequential and adaptive pathways, can be represented without specialty-specific redesign;
- existing catalogues can be migrated without material information loss;
- clinicians and managers can understand the human-readable catalogue record;
- architects can relate pathways to services, capabilities, information dependencies and implementations without embedding technology choices in pathway definitions;
- variants and local implementations can be compared reliably;
- machine processing does not depend on uncontrolled narrative for structural meaning;
- pathway definitions remain stable business artefacts as technology implementations and modelling notations change.

K.4 Principal changes in WD 0.4

Table K.2: Principal changes in WD 0.4

Change	Rationale	Evidence
Scope clarified (Chapter 1): business-level behaviour; computable models, modelling methods and service-subject processes excluded	The boundary with process, case and decision formalisms was implicit and liable to erode.	HL7 Field Guide review; frailty catalogue
FormalRepresentation added (8.10, 3.18)	Platform-independent formal models had no correct place: they are not local implementation artefacts.	HL7 Field Guide review
ImplementationArtefact.localisesRepresentation added	Links a localised model to the shared representation it came from.	HL7 Field Guide review
Unordered stages (8.8, 9.4, 9.6, V56)	Adaptive care could not meet reachability rules without inventing sequences.	Frailty catalogue
accountableFor relationship; coordination role category	Coordination accountability had no construct.	Frailty catalogue; Field Guide responsibility assignment
Pathway-level transitionsTo, escalatesTo at Level 1 (V55); V32 extended to Level 1	Catalogues record relationships between whole pathways before exits and events are authored.	Frailty catalogue
rights attribute on HealthcarePathway and PathwayCatalogue (V57)	Cross-organisational reuse and specialisation require stated reuse conditions.	Field Guide asset metadata

Change	Rationale	Evidence
Task entity, term and V19 withdrawn; term 3.18 reassigned	No conformance level required it, no catalogue used it, and it invited operational detail into definitions.	Necessity review
Principal stages changed from required to recommended (5.3)	Imposed stages were artificial for adaptive pathways.	Frailty catalogue
Population attributes reduced from six to three	Demographic, clinical, risk and eligibility criteria overlapped with inclusion criteria.	Coherence review
Relationship name made optional (8.3)	Naming every edge added burden without value.	Implementability review
InteroperabilityRequirement: realisationHint and assuranceRequirement removed; securityRequirement replaced by accessRequirement	Realisation and assurance belong to implementations; access requirements are expressed as constraints.	Technology neutrality review
Constraint type information-governance	Consent, capacity and proportionate sharing had no precise type.	Frailty catalogue
DecisionPoint.sharedDecision; decision outcomes as permitted outcome set (8.11)	Shared decision-making is a recurrent pathway feature; changes to outcomes need to be governed.	Frailty catalogue; Field Guide design-by-contract principle
Design principles aligned with the project principle set (Table 4)	Minimum sufficient structure and progressive formality were not stated.	Coherence review
Value set additions: entry types case-finding and scheduled-review; care home setting; equity measure domain; social-context concept category; shared-record interaction pattern	Needed to describe neighbourhood and continuing care without local terms.	Frailty catalogue
Consumer requirement 5.6 g) and conformance statement in 5.1	Consumers must not treat representations or conformance as authority.	Authority separation principle
Appendix D extended with HL7 Field Guide positioning and correspondence; Appendix H.2 catalogue findings; Appendix I examples added	Record standards alignment and test evidence.	Development rules 3, 4 and 7
Schema identifier urn:hpds:schema:0.4	New working draft.	–

The changes made in WD 0.5 are listed in Appendix L.

K.5 Decision register

Table K.3: Significant decisions in WD 0.4 and WD 0.5

Decision	Options considered	Rationale
Relationship with the HL7 Field Guide: remain independent; reference and map	Adopt the Field Guide as the HPDS behavioural layer; profile it; reference and map	The Field Guide is an informative modelling method oriented to executable models. Adopting it would make HPDS a workflow modelling specification and tie definitions to three notations. Referencing its outputs keeps HPDS technology neutral and lets organisations that use it gain precision.
Formal representations distinct from implementation artefacts	Reuse ImplementationArtefact; embed models; add a separate reference type	Shared platform-independent models are not local implementations. Embedding would make HPDS depend on external notations. A separate typed reference is the smallest sufficient construct.
Unordered stages rather than a case-management construct	Adopt CMMN concepts (sentries, milestones, required items); add a stage ordering flag; leave as narrative	A single ordering attribute solves the observed problem without importing execution semantics. Milestones are covered by outcomes and completion conditions.
accountableFor rather than full RACI	Full RACI; single coordination relationship; extension	Only accountability was evidenced as missing. Informed parties are covered by InformationProduct.recipient; consulted parties have not been shown to be needed.
No capability construct	Add capabilityRequirement; extension only; derive by analysis	Capability content in the catalogue restated activities and leaked implementation terms. Capability modelling belongs to enterprise architecture (Appendix E).
Governance processes outside scope	Admit service-subject processes as pathways; separate artefact class; exclude	Admitting them would blur the population, entry and outcome semantics on which comparison depends.
Task withdrawn	Retain; make profile-only; withdraw	Unused, unrequired and a route for operational detail. Stages and sub-pathways provide decomposition.
Stable validation rule identifiers	Renumber; retain with withdrawn markers	Catalogues and tools already cite rule numbers; renumbering would break them. This applies HPDS's own identifier principle to itself.
Principal stages recommended at Level 1	Keep mandatory; make optional; recommend	Stages remain valuable where they exist; mandating them produced artificial content.

Decision	Options considered	Rationale
WD 0.5: separate governance state from content	Allow lifecycle edits in place; issue a new version for every status change; governance events	In-place edits break immutability; new versions for status changes break references from implementations and instance data. Appending governance events keeps content immutable and makes current and historical state determinable.
WD 0.5: withdraw splitRule	Retain single and all; retain all only; withdraw	The value single duplicated DecisionPoint without stating who chooses. With it withdrawn, several outgoing precedes always mean concurrency and alternatives always use a decision point.
WD 0.5: sub-pathways placed through activities	Allow precedes to a sub-pathway; activity includes; leave unplaced	An including activity gives the sub-pathway a position in the graph without copying it and without a new relationship type.
WD 0.5: initial nodes rather than a stage start relationship	New startsAt from stages; initial-node rule	The rule needs no new relationship type and supports both sequential and adaptive stages. Ambiguous parallel starts are flagged for review (V64).
WD 0.5: discrepancy structure for formal representations	Free text in note; overload correspondence; structured discrepancies	Structured entries with status let consumers determine what they may rely on while a discrepancy is open.
WD 0.5: HL7 document conventions	Retain ISO clause format; adopt HL7 conventions	HL7 conventions provide upper-case conformance verbs and numbered conformance statements that support traceability. Chapter numbers 2 to 19 are retained to preserve traceability to WD 0.4.

K.6 Open issues

Open issues are listed in Appendix N.

K.7 Earlier history

WD 0.2 introduced the catalogue-centred scope, the ISO 13940:2026 reference, four conformance levels, separation of authority from lifecycle and intended use, removal of implementation from the definition lifecycle, separation of structural and behavioural relationships, and expression profiles. WD 0.3 restored structured data types, relationship attributes and cardinalities, sub-pathway inclusion, branch rejoin, conditional attribute conditions, intended-use constraints, assurance evidence, instance reference points and the validation rules omitted from WD 0.2. Details are recorded in Annex K of WD 0.3.

Appendix L Change log from WD 0.4

(informative)

Table L.1 lists the changes made in WD 0.5, grouped by the review finding that prompted them. Each finding was verified against the WD 0.4 text before a change was made. Where WD 0.4 already contained a partial solution, the change extends that provision rather than introducing a second mechanism.

Table L.1: Changes in WD 0.5

Issue	Affected provisions	Change	Rationale
1 Boundary of behavioural semantics	1.2; 9.7; 9.8 (new); 3.59; Table 47 (precedes)	Scope now states that HPDS defines descriptive behavioural semantics and lists the analyses they support; execution semantics are enumerated and excluded. New 9.8 gives the descriptive meaning, business purpose and supported analyses of each behavioural construct.	WD 0.4 excluded all control-flow semantics while specifying splits, joins, branching, iteration and reachability.
1 Behavioural constructs without a determinate meaning	Table 46 (splitRule withdrawn); Table 48; V24; Appendix D	splitRule withdrawn. Several outgoing precedes always denote concurrency; alternatives use a DecisionPoint. Deferred selection defined.	The value single duplicated DecisionPoint without stating who chooses, so two implementers could read the same graph differently.
2 Ordered-stage entry and completion	8.8; 3.60; 9.4; 9.6; V64	Initial nodes defined; they commence when an ordered stage commences. Terminal nodes and completion defined. Multiple initial nodes flagged for review.	WD 0.4 did not say how contained nodes of an ordered stage become reachable.
2 Unordered and nested stages, optional activities	8.8; 9.6	Reachability rules c) and d) cover unordered and nested stages; optional activities are reachable and affect completion only.	Completes WD 0.4 9.6 a).
2 Sub-pathway inclusion in behavioural analysis	Table 47; 8.9; 9.3; 9.4; V60	includes may have an Activity source; at Level 2 it must. The including activity is a single, opaque behavioural node.	WD 0.4 gave an included sub-pathway no position in the graph.
2 External events used as anchors	8.13; 3.23; 9.3; 9.6; V18; V28	External events have no container, take no part in reachability unless they have outgoing behavioural relationships, and need no effect when used only as anchors.	WD 0.4 required a container for every event and exempted external events from reachability only when their effect was initiates.
2 Cycle detection	9.5; V30; V68	Cycle-detection graph defined, including selects and implicit stage edges; iterates excluded and checked separately.	In WD 0.4 a cycle through a decision point and its branches escaped V30.
2 Invalid, incomplete and review outcomes	9.9 (new); 5.7; Table 50	Outcomes by level stated.	WD 0.4 relaxed graph rules at Level 1 only implicitly.
3 Named organisations	6.1; V09; V54; J5	Prohibition narrowed to implementation-specific delivery arrangements in care-model content; named organisations permitted in listed governance and provenance metadata, including those of formal representations.	Under WD 0.4 a definition could fail V09 because its mandatory owner was a named organisation.
4 Definition-level deviations	8.4 (deviation); 13.2; V61	Deviations stored in HealthcarePathway.deviation; completeness defined by identifier comparison with the parent version.	13.2 required deviations but the model had nowhere to hold them.
4 Implementation clinical authority	8.27; 16.1; V06	PathwayImplementation.clinicalAuthority added; approval of the realised definition does not transfer.	Table 54 and V06 required clinical authority that Table 43 could not record.
4 Concepts in criteria	8.26.2 (Expression.concept); 11.3; V39	Expressions identify the concepts they depend on; Level 3 binds them.	V39 assumed concepts could be inferred from narrative.
4 Entities and structured data types	8.1; 8.2; 8.3; 7.2; Table 7	Entity types and structured data types listed; only entities carry common attributes; referenceable data types identified.	CatalogueEntry and FormalRepresentation were ambiguous.
5 Immutability and lifecycle	14.1; 14.2; 14.5; 14.6 (new); 8.4; 8.27; 8.26.10 (GovernanceEvent); V14; V62; V63	Governance-state attributes separated from content and changed only by appended governance events; historical and current state defined.	14.2 forbade changes that 14.5 required.
6 Level 4 coverage	5.2; 8.11; 13.7 (new); Table 52; V42; V44	Every expression-bearing attribute listed with its Level 4 obligation; judgement exception limited to the decision criteria and branch conditions of the same decision point; authorised exceptions defined.	WD 0.4 left population criteria, branch conditions and exceptions ambiguous.

Issue	Affected provisions	Change	Rationale
7 Authority of formal representations	8.10; 8.10.1 to 8.10.3 (new); 8.26.11 (RepresentationDiscrepancy); 19.3; 5.6 g); V66	Differences classified as contradiction, elaboration, incomplete coverage or unresolved; contradictions and unresolved discrepancies recorded with status; reliance while open defined; source correspondence handled.	WD 0.4 treated every difference as a defect in the representation and misused the correspondence attribute.
8 Unsupported modifying extensions	13.6; 5.6 d); J14	Affected content defined; non-interpretation at every level; storage, display, preservation and reporting specified.	WD 0.4 restricted interpretation only for Level 4 purposes.
8 Level 3 transparency	11.5 (new); 5.1; 5.6 h); Table 1; J24	Binding report required with Level 3 and 4 claims; what Level 3 demonstrates stated.	Local terms satisfied Level 3 invisibly.
9 Adoption burden	5.7 (new); Chapter 17; Appendix G	Obligations by level stated; Appendix G made a recommended reference presentation; mandatory visibility and source identification kept.	The fixed order of Appendix G was not defensible for tabular and application views.
10 Implementation readiness	1.9; 19.1; Appendices B, I, J, K, N	Claims limited to artefacts produced and tests run; schema, validator, examples and tests provided and their results reported; missing evidence listed.	WD 0.4 described a schema and metamodel source that did not exist.
11 Consistency: attribute obligations not validated	8.1; V59	General rule for M and C obligations and multiplicities; default for absent optional Booleans.	No WD 0.4 rule covered most attribute obligations.
11 Consistency: value sets	8.2; 13.3; B.2 g)	Domain profiles may restrict but not extend value sets.	8.2 and B.2 permitted extension that 13.3 forbade.
11 Consistency: extension relationship types	9.1	Extension relationship types identified by absolute URIs.	Needed for schema validation.
11 Consistency: rule scope	V17, V32, V42	V17 and V42 apply from Level 1; V32 wording retained.	Present relationships and expressions must be well formed at every level.
11 Consistency: unverified claims	Chapter 2 NOTE; F.1; 1.5; B.1	ISO 13940:2026 status, mappings and schema testing stated as unverified where they are.	Claims of alignment must not exceed the evidence.
Format	Whole document	HL7 specification conventions: document information, revision history, Introduction chapter, upper-case conformance verbs, numbered conformance statements, appendices. Chapter numbers 2 to 19 retained.	Requested format; retaining numbers preserves traceability to WD 0.4.

Appendix M Requirements traceability

(informative)

M.1 Method

Every normative statement containing **SHALL** or **SHALL NOT** carries a conformance statement identifier (1.8.2). Table M.1 relates each requirement revised or added in WD 0.5 to its representation in the model and to the validation rules that test it. Table M.2 is generated from the text and the rule set. It lists every conformance statement with the rules whose source provisions (Table A.1) include the statement's section. A statement with no rule is either tested by Review, applies to a conformance claim or a consumer rather than a description, or has no automated test; such statements are counted in Appendix N.

M.2 Revised and new requirements

Table M.1: Traceability of revised and new requirements

Provision	Requirement	Model representation	Validation
1.2, 9.8	Behavioural relationships have the descriptive semantics of Table 49; execution semantics excluded.	Relationship; Table 47	V17 to V33, V55, V60, V64, V65, V67, V68
5.7, 9.9	Obligations and graph outcomes by level.	Conformance level (HealthcarePathway.conformanceLevel)	Applies column of Table A.1
6.1	No implementation-specific delivery arrangements in care-model content; named organisations permitted in governance metadata.	Care-model entities; governance attributes listed in 6.1	V09, V54 (Review)
7.2, 8.1	Entity types carry common attributes; structured data types do not.	Table 7; Table 9; Appendix B collections	V01, V59 (schema)
8.1	M and C obligations, multiplicities and Boolean defaults.	All attribute tables	V59
8.4, 13.2	Variant deviations recorded and complete.	HealthcarePathway.deviation; Deviation	V37, V61
8.8, 9.6	Ordered-stage initial nodes; unordered-stage availability; completion.	PathwayStage.ordering; PathwayStage.completionCondition	V28, V56, V64
8.10.1 to 8.10.3	Contradictions and unresolved discrepancies recorded; reliance while open.	FormalRepresentation.discrepancy; RepresentationDiscrepancy	V66; consumer requirement 5.6 g)
8.11, 13.7	Judgement exception scope.	DecisionPoint.judgementBased; PathwayBranch.condition	V44
8.13, 9.3	External events have no container and need no effect when used only as anchors.	Event.external; Event.effect	V18, V28, V59
8.26.2, 11.3	Concepts on which criteria depend are identified and, at Level 3, bound or local.	Expression.concept; ClinicalConcept; TerminologyBinding	V38, V39
8.27, 16.1	Implementation has its own clinical authority; no transfer from definition.	PathwayImplementation.clinicalAuthority; GovernanceEvent	V06
9.1	splitRule withdrawn; join rules consistent.	Relationship.joinRule	V24
9.2, 9.3	includes from an Activity at Level 2; sub-pathway opaque.	Relationship (includes)	V17, V60
9.5	Cycle-detection graph acyclic; iterates meaningful.	Relationship; implicit stage edges	V30, V68
9.6	Reachability, termination, waitsFor targets.	Relationship; PathwayStage.ordering; Event.external	V28, V29, V65, V67
11.5, 5.1	Binding report with Level 3 and 4 claims.	Conformance claim	J24 (claim review); validator report
13.6	Unsupported modifying extensions not interpreted; preserved; disclosed.	Extension.modifying; modifyingExtension	V13; consumer requirement 5.6 d)
13.7	Level 4 coverage of expression-bearing attributes.	Table 52 attributes; Expression	V42, V44, V45
14.2	Content immutable; governance state changed only by appended events.	Governance-state attributes; GovernanceEvent	V14, V62, V63
14.6	Historical and current state determinable.	GovernanceEvent	V62; consumer requirement 5.6 i)
Chapter 17, Appendix G	Mandatory visibility; reference presentation recommended.	Presentation (not serialised)	J19 (Review)

Provision	Requirement	Model representation	Validation
19.3	Conceptual model prevails over serialisation, schema and mapping.	Appendix B	J20

M.3 Conformance statements and validation rules

Table M.2: Conformance statements, model elements and validation rules

Statement	Section	Statement text	Model elements named	Validation rules
CONF-HPDS-001	1.8.1	SHALL and SHALL NOT indicate a requirement that is necessary for conformance;	–	None
CONF-HPDS-002	1.8.2	Each normative statement containing SHALL or SHALL NOT is labelled with a conformance statement identifier of the form CONF-HPDS-nnn. The identifiers are assigned in docu...	–	None
CONF-HPDS-003	5.1	Conformance MAY be claimed by a pathway definition, a pathway implementation, a pathway catalogue, a pathway description producer or a pathway description consumer. A cla...	–	None
CONF-HPDS-004	5.1	A conformance claim concerns the representation of a pathway only. It SHALL NOT be presented as, or used to imply, clinical approval, authority or fitness for a particula...	–	None
CONF-HPDS-005	5.2	A pathway definition SHALL declare exactly one conformance level from Table 1. Each level includes all requirements of the levels below it.	–	None
CONF-HPDS-006	5.3	Every pathway definition SHALL contain the items in Table 2, irrespective of conformance level.	–	V02, V03
CONF-HPDS-007	5.4	A conformant pathway implementation SHALL reference one pathway definition and its version, identify its implementing organisation or organisations, map relevant definiti...	–	V46, V48
CONF-HPDS-008	5.5	A conformant pathway catalogue SHALL meet the following requirements:	–	V50, V51, V52
CONF-HPDS-009	5.5	each catalogue entry SHALL reference a conformant pathway definition by identifier and version;	–	V50, V51, V52
CONF-HPDS-010	5.5	catalogue entry identifiers SHALL be unique within the catalogue and SHALL remain stable across catalogue releases;	–	V50, V51, V52
CONF-HPDS-011	5.5	catalogue metadata SHALL identify publisher, governance authority, scope and catalogue version;	–	V50, V51, V52
CONF-HPDS-012	5.5	catalogue search and export SHALL preserve pathway identifiers and versions;	–	V50, V51, V52
CONF-HPDS-013	5.5	a catalogue SHALL NOT represent implementation-specific content as part of a pathway definition;	–	V50, V51, V52
CONF-HPDS-014	5.5	all validation rules designated for catalogues in Appendix A SHALL be met.	–	V50, V51, V52
CONF-HPDS-015	5.6	A conformant pathway description producer SHALL :	–	None
CONF-HPDS-016	5.6	A conformant pathway description consumer SHALL :	–	None
CONF-HPDS-017	6.1	Whether a pathway description describes a pathway definition or a pathway implementation is declared by its root element: HealthcarePathway for a definition and PathwayIm...	HealthcarePathway; PathwayImplementation	V09, V54
CONF-HPDS-018	6.1	A pathway definition SHALL describe the intended model of care without implementation-specific delivery arrangements. Implementation-specific delivery arrangements are: n...	PathwayImplementation	V09, V54
CONF-HPDS-019	6.1	The same separation applies to formal representations. A formal representation referenced by a pathway definition (8.10) SHALL contain no implementation-specific delivery...	–	V09, V54
CONF-HPDS-020	7.1	A PathwayCatalogue is a governed collection of pathway definitions. A catalogue manifest SHALL reference its entries by identifier and version and SHALL NOT embed or dupl...	PathwayCatalogue	V52

Statement	Section	Statement text	Model elements named	Validation rules
CONF-HPDS-021	7.3	A conformant catalogue SHOULD support discovery using pathway name, identifier, pathway type, population, owner, service, care setting, intended use, authority status, li...	–	None
CONF-HPDS-022	8.3	Every element of an entity type listed in Table 7, including every Relationship (9.1), SHALL carry the common attributes in Table 9 in addition to the attributes specifie...	–	V01
CONF-HPDS-023	8.4	HealthcarePathway is the root business artefact of a pathway definition and the principal unit referenced by a pathway catalogue. A pathway definition SHALL contain exact...	HealthcarePathway	V05, V08, V10, V11
CONF-HPDS-024	8.5	A pathway definition SHALL contain at least one Population element. Population defines the set of subjects of care to whom the pathway applies.	–	V03
CONF-HPDS-025	8.6	A pathway definition SHALL contain at least one EntryPoint element. The entry point records the route of arrival and the entry criteria record the conditions that SHALL h...	EntryPoint	V03, V20, V31
CONF-HPDS-026	8.7	A pathway definition SHALL contain at least one ExitPoint element. A pathway definition SHOULD represent every exit type in Table 13 that can apply to the pathway, not on...	ExitPoint	V03, V31
CONF-HPDS-027	8.8	A stage is complete when its completion conditions hold. Where an ordered stage has no completion condition, it is complete when each of its terminal nodes that is not an...	–	V56, V64
CONF-HPDS-028	8.10	a formal representation SHALL be technology neutral in the sense of 6.1; a model localised to particular organisations, locations, applications, interfaces or local proce...	ImplementationArtefact	V09, V53, V54
CONF-HPDS-029	8.10	a formal representation SHALL NOT be the only place in which an element or relationship required for the declared conformance level is recorded;	–	V09, V53, V54
CONF-HPDS-030	8.10	differences between a formal representation and the pathway definition SHALL be classified and handled as specified in 8.10.1 to 8.10.3;	–	V09, V53, V54
CONF-HPDS-031	8.10.1	Elaboration and incomplete coverage are permitted and are not defects; they SHOULD be explained in the note component. Contradictions and unresolved discrepancies SHALL b...	RepresentationDiscrepancy	V09, V53, V54, V66
CONF-HPDS-032	8.10.2	Where correspondence is source, a contradiction indicates an error in the derivation of the definition or a later change to the representation. It SHALL NOT be resolved b...	–	V09, V53, V54
CONF-HPDS-033	8.10.3	a consumer SHALL NOT rely on the formal representation;	–	V09, V53, V54, V66
CONF-HPDS-034	8.10.3	a presentation of the formal representation, or of the affected element, SHALL indicate that an open discrepancy exists.	–	V09, V53, V54, V66
CONF-HPDS-035	8.11	A DecisionPoint is a point at which one or more alternative pathway branches are selected. Responsible roles are recorded by decides relationships. Where selection relies...	DecisionPoint	V21, V26, V45
CONF-HPDS-036	8.11	The branches of a decision point define its permitted outcomes. A pathway variant or pathway implementation that adds, removes or substitutes an outcome SHALL record the ...	–	V21, V26, V45
CONF-HPDS-037	8.13	An Event is a clinically or operationally relevant occurrence that can influence pathway behaviour. An occurrence outside the pathway that serves as an anchor for a tempo...	–	None
CONF-HPDS-038	8.15	A Service is a healthcare or care service that delivers or supports pathway elements. A Service in a pathway definition SHALL NOT identify a named provider organisation. ...	–	None
CONF-HPDS-039	8.17	An InformationRequirement specifies information needed to support an activity or decision. The elements that need it are identified by requires relationships. Every infor...	InformationRequirement	V27

Statement	Section	Statement text	Model elements named	Validation rules
CONF-HPDS-040	8.20	An Outcome is an intended result of the pathway or of a component of it. A pathway definition SHALL contain at least one Outcome. An Outcome SHALL NOT be defined solely a...	–	V03
CONF-HPDS-041	8.23	Exceptions SHALL be represented as Exception elements and SHALL NOT be recorded solely within narrative attributes of other elements.	–	None
CONF-HPDS-042	8.27	A PathwayImplementation describes how an organisation or health system realises a pathway definition. It is not part of the technology-neutral pathway definition. An impl...	PathwayImplementation	V06, V46, V47
CONF-HPDS-043	9.1	Relationships SHALL be represented explicitly as identified Relationship elements carrying the common attributes in 8.3 and the attributes in Table 46. Behaviour SHALL NO...	–	V17, V24
CONF-HPDS-044	9.2	At Level 2 and above, the source of every transitionsTo and escalatesTo relationship SHALL be an ExitPoint, Event or DecisionPoint (V55), and the source of every includes...	DecisionPoint; ExitPoint	V17, V20, V21, V22, V26, V32, V55, V60
CONF-HPDS-045	9.3	At Level 2 and above, every PathwayStage, Activity and DecisionPoint, and every Event whose external attribute is not true, SHALL have exactly one structural container wi...	DecisionPoint; PathwayStage	V18, V60
CONF-HPDS-046	9.3	A sub-pathway SHALL be included by an includes relationship and SHALL NOT be copied into the including definition. The relationship SHALL identify the version of the sub-...	–	V18, V60
CONF-HPDS-047	9.4	The behavioural patterns in Table 48 SHALL be represented as specified.	–	None
CONF-HPDS-048	9.5	Repetition SHALL be represented only by iterates relationships, each carrying an IterationSpec with a termination condition. The source of an iterates relationship is the...	IterationSpec	V30, V68
CONF-HPDS-049	9.5	Iterates, interrupts, waitsFor and non-behavioural relationships are not edges of the cycle-detection graph. The cycle-detection graph SHALL be acyclic (V30). The source ...	–	V30, V68
CONF-HPDS-050	9.6	every behavioural node other than an Event whose external attribute is true SHALL be reachable (V28);	–	V28, V29, V64, V65, V67
CONF-HPDS-051	9.6	for each EntryPoint, applying the rules above with only the startsAt relationship of that entry point in rule a), at least one ExitPoint SHALL be reachable (V29);	EntryPoint; ExitPoint	V28, V29, V64, V65, V67
CONF-HPDS-052	9.6	the target of every waitsFor relationship SHALL be an Event whose external attribute is true or a reachable Event (V67).	–	V28, V29, V64, V65, V67
CONF-HPDS-053	9.9	For a declared conformance level, a behavioural graph is invalid if any validation rule of severity Error that applies at that level fails. At Level 1 the graph MAY be in...	–	V17, V65
CONF-HPDS-054	10.1	Temporal constraints SHALL be represented using the TemporalConstraint structure in Table 51. Every temporal constraint SHALL have a narrative statement. At Level 4, temp...	TemporalConstraint	V35
CONF-HPDS-055	10.2	Temporal constraints that depend on a prior occurrence SHALL be expressed relative to an anchor rather than as calendar dates.	–	None
CONF-HPDS-056	10.3	A pathway definition SHOULD NOT contain temporal constraints that cannot be satisfied together. A producer that detects such constraints SHALL report a warning (Appendix ...	–	V36
CONF-HPDS-057	11.2	A terminology binding SHALL conform to 8.26.1 and SHALL identify the semantic resource, its version and the code, value set or data element identifier. Value set bindings...	–	V38
CONF-HPDS-058	11.3	Local terms SHALL be identified by setting localTerm to true and SHALL NOT be represented as codes from an external semantic resource. At Level 3 and above, the clinical ...	ClinicalConcept; InformationRequirement	V39

Statement	Section	Statement text	Model elements named	Validation rules
CONF-HPDS-059	11.4	A change to the version of a bound semantic resource SHALL be recorded in the change record of the version that introduces it. Producers SHOULD report bindings to version...	–	V40, V41
CONF-HPDS-060	11.5	A Level 3 or Level 4 conformance claim SHALL include a binding report. For the ClinicalConcepts in scope of V39 the report SHALL state: the number bound to external seman...	ClinicalConcepts	V39
CONF-HPDS-061	12	An InteroperabilityRequirement (8.25) states a business information exchange on which a pathway depends. It SHALL identify the information, sender, receiver, trigger and ...	InteroperabilityRequirement	V34
CONF-HPDS-062	13.1	A pathway definition MAY specialise another pathway definition, and specialisation MAY be repeated: a national or professional reference pathway MAY be specialised region...	–	V08
CONF-HPDS-063	13.2	An inherited element that retains the same meaning SHALL retain its parent identifier. A new element, or an inherited element whose meaning changes materially, SHALL rece...	HealthcarePathway	V37, V47, V61
CONF-HPDS-064	13.3	A domain profile MAY tighten cardinalities and obligations, restrict value sets, require particular bindings and define extensions. It SHALL NOT relax any requirement of ...	–	None
CONF-HPDS-065	13.4	A Level 4 conformance claim SHALL identify at least one expression profile. An expression profile SHALL define the expression language and version, the evaluation context...	–	V42, V43
CONF-HPDS-066	13.5	A domain profile or expression profile SHALL have a persistent identifier, a version and a statement of the version of this document on which it is based.	–	None
CONF-HPDS-067	13.6	Extensions SHALL conform to 8.26.9. A non-modifying extension SHALL NOT change the interpretation of core content. Every modifying extension used in a description SHALL b...	HealthcarePathway; PathwayImplementation	V13
CONF-HPDS-068	13.6	SHALL NOT interpret affected content for any purpose that depends on its meaning, at any conformance level, including semantic validation, comparison, analysis, derivatio...	–	V13
CONF-HPDS-069	13.6	SHALL preserve the extension and the affected content without alteration when storing or re-serialising the description;	–	V13
CONF-HPDS-070	13.6	SHALL report the unsupported modifying extensions and the identifiers of the affected elements whenever it reports validation or analysis results.	–	V13
CONF-HPDS-071	13.7	Table 52 lists every attribute whose type is Expression, or which contains an Expression, and states whether a computable expression is required at Level 4. At every leve...	DecisionPoint	V44
CONF-HPDS-072	14.1	Every pathway definition and pathway implementation SHALL record owner, version, lifecycle status, authority status and intended use. From the published state onward, eff...	ApprovalRecords	V05, V11, V12, V57
CONF-HPDS-073	14.2	Once a pathway description has been published, its content SHALL NOT be modified; any change to content SHALL produce a new version (Chapter 15). Its governance state MAY ...	GovernanceEvent; RepresentationDiscrepancy	V06, V14, V62, V63
CONF-HPDS-074	14.2	A GovernanceEvent of type publication SHALL be recorded when a description is published; it establishes the lifecycle status, authority status and review date at publicat...	GovernanceEvent	V06, V14, V62, V63
CONF-HPDS-075	14.2	Historical versions SHALL remain interpretable and SHALL remain retrievable while they are referenced by implementations, catalogues or other pathway definitions. Catalog...	–	V06, V14, V62, V63
CONF-HPDS-076	14.3	Provenance SHALL be recorded for the pathway description as a whole and for individual elements whose origin differs materially from that of the whole description. Conten...	ProvenanceRecord	None

Statement	Section	Statement text	Model elements named	Validation rules
CONF-HPDS-077	14.4	Every version after the first published version SHALL carry a ChangeRecord (8.26.7) that states the version, date, author, change category, summary and rationale, identif...	ChangeRecord	V15
CONF-HPDS-078	14.5	Pathway descriptions SHALL use the lifecycle states and permitted transitions in Table 53. Organisations MAY use local lifecycle terminology provided each local state map...	–	None
CONF-HPDS-079	14.6	A consumer SHALL use the current governance state when presenting a description. A consumer SHOULD use the historical governance state when interpreting instance data rec...	–	V06, V12, V62
CONF-HPDS-080	15.1	Persistent identifiers SHALL be assigned to pathway descriptions, catalogue entries, formal representations and all elements, including stages, activities, decision point...	–	V01
CONF-HPDS-081	15.2	An element that retains its meaning in a later version SHALL retain its identifier. An element whose meaning changes materially SHALL receive a new identifier, and the ch...	–	None
CONF-HPDS-082	15.4	A reference to an element in another pathway description SHALL identify that description and its version, or an explicit permitted version range.	–	V16, V32
CONF-HPDS-083	16.1	Every pathway definition and implementation SHALL declare one authority status from Table 54. The additional requirements of Table 54 apply to the current authority statu...	PathwayImplementation	V04, V06
CONF-HPDS-084	16.2	A pathway description SHALL declare one or more intended uses from Table 55. Intended use is independent of authority status, but the combinations in the third column are...	–	V04, V07
CONF-HPDS-085	16.2	For the purpose of the third column, authority statuses are ordered as listed in Table 54, from unapproved to operationally-authoritative. Consumers SHALL display authori...	–	V04, V07
CONF-HPDS-086	16.3	This document does not define an executable authority status. Where pathway content is used to configure or drive automated clinical or operational activity, the pathway ...	ImplementationArtefact	V49
CONF-HPDS-087	17	Every pathway definition SHALL be capable of presentation as a human-readable catalogue record. Every presentation of a pathway definition, in any view, SHALL display its...	–	None
CONF-HPDS-088	17	Tabular and graphical views, including diagrams generated from formal representations, are projections of the underlying pathway definition. They SHALL identify the sourc...	–	None
CONF-HPDS-089	18	At Level 2 and above, a pathway definition SHALL be transformable without semantic loss into a directed labelled graph in which elements are nodes and relationships are e...	–	None
CONF-HPDS-090	19.2	Other serialisations, such as JSON-LD [11], MAY be used where they preserve all information and identifiers and can be transformed to and from the normative serialisation...	–	None
CONF-HPDS-091	19.3	Where a serialisation, schema or mapping conflicts with Chapters 7 to 16, the conceptual model prevails and the inconsistency SHALL be treated as a defect in the serialis...	–	None
CONF-HPDS-092	A.1	Pathway description producers SHALL apply the rules in Table A.1. A rule of severity Error prevents a conformance claim for the levels or classes shown. A rule of severit...	–	None
CONF-HPDS-093	A.1	The Method column states how a rule is evaluated: Structural (within one description, by schema or simple checks); Conditional (attribute conditions within one descriptio...	–	None
CONF-HPDS-094	B.2	The serialisation SHALL meet the following requirements:	–	None

Statement	Section	Statement text	Model elements named	Validation rules
CONF-HPDS-095	B.2	undeclared properties are not permitted; additions SHALL use the extension mechanism (13.6);	–	None
CONF-HPDS-096	J	A conformance claim SHALL be supported by a completed checklist in the form of Table J.1. Items that do not apply to the conformance class claimed SHALL be marked not app...	–	None

Appendix N Unresolved issues and missing implementation evidence

(informative)

N.1 Minimum release package

The following release package is the minimum required for meaningful conformance testing. Table N.1 states, for each component, what exists for WD 0.5 and what remains outstanding.

Table N.1: Minimum release package: status

Component	Status	Evidence	Outstanding
Schema aligned with the revised conceptual model	Produced	hpds.schema.0.5.json generated from the attribute tables of this document; checked as a valid JSON Schema Draft 2020-12 schema; all 40 test documents validated with the Python jsonschema library.	Independent review of the schema; testing with a second JSON Schema implementation.
Requirements-to-validation traceability matrix	Produced	Table M.1 (revised requirements) and Table M.2 (all 96 conformance statements), also supplied as traceability.csv. 28 statements have no associated validation rule.	Review of the statements without rules (Appendix N, issue 4).
Graph and cross-version validation rules with testable outcomes	Produced in part	Rules defined in Chapter 9, 13.2 and 14.2. The reference validator implements 60 of 67 active rules; the following are not implemented: V09, V15, V36, V40, V41, V47, V54. Rules requiring review or an external service: V03, V09, V36, V40, V47, V54.	Independent implementation; test pairs for V15, V41 and V51; review criteria.
At least one declared expression profile for Level 4 testing	Outstanding	A structural test fixture declares the content required by 13.4 so that V43 and V44 can be exercised. It defines no evaluation semantics.	An interoperable expression profile with evaluation semantics and test expressions (Appendix N, issue 2).
Complete sequential and adaptive examples from identified sources	Produced in part	Sequential: Hello Patient (HL7 Field Guide, illustrative). Adaptive: FNM-CGA-001 (frailty catalogue). Both at Level 2; Level 3 and Level 4 variants used in tests. Not clinically reviewed.	Examples from published clinical guidance for endoscopy, maternity, pathology, acute admission and a WHO DAK; review of the produced examples.
Valid and deliberately invalid test cases with expected results	Produced	40 test cases (16 valid, 24 invalid or with warnings) exercising 24 rules; the reference validator passes 40 of 40.	Execution by an independent implementation (Appendix N, issue 1); consumer and presentation scenarios.

N.2 Prioritised unresolved issues

The issues below are listed in priority order. Priority 1 issues prevent the acceptance criterion from being demonstrated: that two independent implementers reach the same interpretation of the core requirements and the same conformance result for the supplied test cases.

Table N.2: Unresolved issues and missing evidence

No.	Priority	Issue	Outstanding work
1	1	Only one implementation of the validation rules exists: the reference validator produced with this draft. Agreement between independent implementations has not been tested.	An independent implementer builds a validator from the text alone and runs the Appendix I test cases; differences are resolved in the text.
2	1	No interoperable expression profile exists. The Level 4 test uses a structural test fixture that declares the required content of a profile but defines no evaluation semantics, so computability of expressions has not been tested.	Define at least one expression profile (for example CQL or FHIRPath over a declared data model), with executable test expressions and expected results.
3	1	Rules evaluated by Review (V03 outcome condition, V09, V36, V47, V54) have no written review criteria, so reviewers can reach different results.	Write review criteria and examples for each Review rule; test them with two reviewers on the same descriptions.
4	2	Conformance statements not tested by any rule (see Table N.1 and Table M.2), including consumer requirements (5.6, 8.10.3, 13.6, 14.6) and presentation requirements (Chapter 17).	Define consumer and presentation test scenarios with expected behaviour.
5	2	Cross-version rules (V14, V15, V41, V51) need a version history. The reference validator evaluates V14 and V61 only for pairs supplied in test cases.	Specify how a catalogue or repository supplies version history to validators; add test pairs for V15, V41 and V51.
6	2	ISO 13940:2026: publication and final text not verified; term-by-term reconciliation with Chapter 3 not done.	Obtain the published text; reconcile terms; revise Chapter 2 and Appendix F.

No.	Priority	Issue	Outstanding work
7	2	Worked examples outstanding: endoscopy, maternity, pathology, acute admission and discharge, WHO DAK mapping. The produced examples have not been reviewed by clinicians or by the authors of their sources.	Identify sources; produce examples; obtain review of structure (not clinical approval).
8	3	Mappings in Appendices C to F are not verified by round-trip conversion. The correspondence between conformance levels and FHIR Clinical Practice Guidelines knowledge levels is unconfirmed.	Round-trip the Hello Patient example through BPMN, CMMN and DMN, and a FHIR PlanDefinition; confirm or revise the mappings.
9	3	The PRSB-format information model workbook (version 0.3.0) is aligned with WD 0.4, not WD 0.5.	Regenerate the workbook from the WD 0.5 attribute tables.
10	3	V40 requires a terminology service that reports withdrawn and superseded versions.	Specify the service interface assumed, or reclassify V40 as advisory.
11	3	Detection of differences between formal representations and definitions is manual; there is no automated comparison.	Define the comparison for mapped constructs once mappings are verified (issue 8).
12	3	The semantics of deferred selection and of relationship conditions depend on expression profiles for computable evaluation.	Address in the expression profile (issue 2).
13	4	Carried from WD 0.4: identifier correspondence between definitions and formal representations; consulted and informed responsibilities; required elements in unordered stages; controlled vocabulary for rights; monitoring of OMG shared data and provenance specifications.	Decide after further catalogue evidence.
14	4	Licence terms for this document and the release package have not been set.	Owner decision.
15	4	Frailty catalogue entries lack pathway owners, exit points and a governance authority. The examples use the example author as owner.	Catalogue action, not a model issue.

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